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Project 2

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Abstract

This project focuses on the design and implementation of a multi-area campus network using Cisco Packet Tracer. A single IPv4 base network, 102.71.128.0/17, was used as the foundation for the addressing scheme. Variable Length Subnet Masking (VLSM) was applied to divide the network into multiple LAN and WLAN segments according to host requirements while minimizing IP address waste. Static IP addresses were assigned to routers and end devices, and /30 subnets were used for router-to-router point-to-point links to optimize address utilization. Dynamic routing was implemented using the Open Shortest Path First (OSPF) protocol with a hierarchical multi-area design, where Area 0 acted as the backbone and Areas 20 and 30 represented the West and East networks. Network functionality was verified through interface status checks, OSPF neighbor validation, routing table inspection using show ip route, and end-to-end connectivity testing using ping and traceroute across both wired and wireless segments. The results demonstrate that the implemented network design meets the project requirements and supports reliable communication across all network areas.

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Theory:

IP Addressing and Subnetting

IP addressing and subnetting are fundamental concepts in computer networking that enable devices to communicate efficiently within a network. An IP address provides a unique identifier for each device, allowing data packets to be routed accurately to their intended destinations. Without IP addressing, communication between devices on a network would not be possible. [1]

IP Addressing

Each device that uses a network receives an IP address and a special identifier number. IP Addresses are necessary for routing data packets between devices and enabling Internet communication between devices. Dotted decimal notation, which depicts four sets of bit. The network, separated by periods, is the most common way to express IP addresses. The address is 32 bits, and every number corresponds to a byte of the address.

There are two primary forms of IP addresses IPv4 and IPv6. Because of the 32-bit length and the limited amount of unique IPv4 addresses, subnets and various methods for storing IP addresses have been developed. There are many more unique addresses available for IPv6 addresses that are 128-bit.[2]

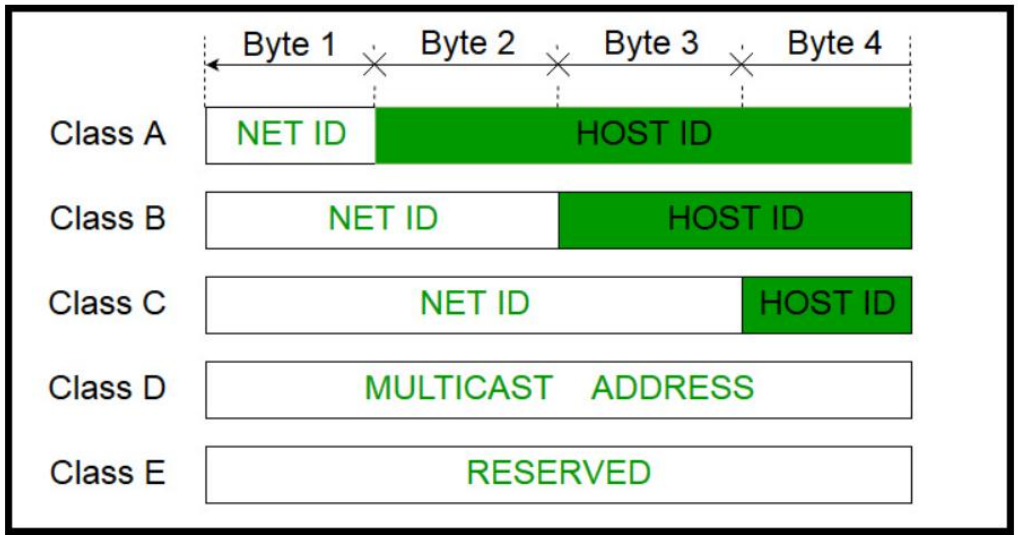


Figure 1:IP Addressing[2]

Subnetting

A network is divided into smaller subnetworks, or subnetworks, through the process known as a subnetwork. For this purpose, a network part of the IP address is created by taking bits from the host part. The host party identifies the specific device on the subnetwork, while the network party identifies the subnetwork as a whole.

The sub-network enables network managers to create more controllable and segmented networks for performance or security needs. For example, a large enterprise could segment its network into subnetworks for multiple divisions or locations.

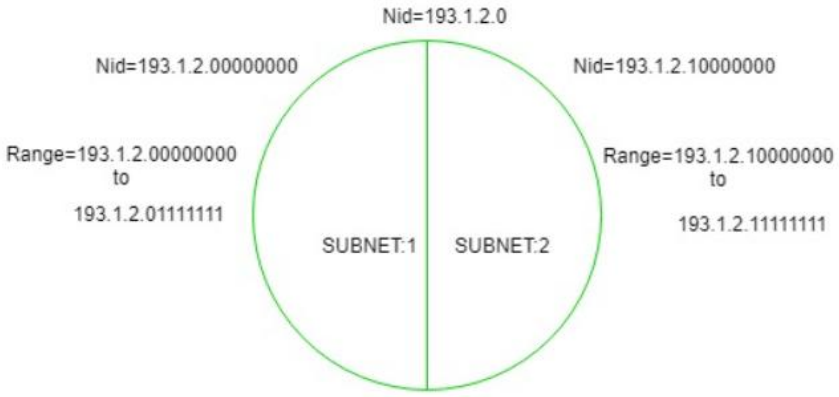


Figure 2:Subnetting[2]

Subnetting Is Used in Large Networks

Subnetting is used in large networks to divide a single large network into smaller, more manageable subnetworks. This reduces network congestion by limiting broadcast traffic within each subnet, improves performance, and makes network management easier. Subnetting also enhances security by isolating different parts of the network and allows more efficient utilization of IP addresses.

CIDR Notation (/23, /24, /30)

CIDR (Classless Inter-Domain Routing) notation is used to represent the subnet mask and define the size of a network. The number following the slash indicates how many bits are used for the network portion of the IP address. For example, a /24 network provides more host addresses than a /30 network. CIDR allows flexible subnet sizes and supports efficient IP address allocation without relying on traditional IP classes.

Difference Between Network Address, Broadcast Address, and Usable IP Addresses

The **network address** identifies the subnet itself and cannot be assigned to any device. The **broadcast address** is used to send data to all devices within the same subnet and is also not assignable. **Usable IP addresses** are the remaining addresses between the network and broadcast addresses, which can be assigned to hosts such as computers, routers, and other network devices.

Network Topology Design:

A network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them. There are two main types of topology:

- **Physical Topology:** The actual physical layout of cables and devices.
- **Logical Topology:** How data moves across the network, regardless of physical layout.[3]

Types of Network Topologies

Several types of network topologies are commonly used in computer networks, each defining a specific way in which devices are connected and communicate. Common topology types include **Point-to-Point**, **Star**, **Bus**, **Ring**, **Mesh**, **Tree**, and **Hybrid** topologies. Each topology has its own characteristics and is selected based on factors such as network size, performance requirements, reliability, cost, and ease of expansion.[3]

Hybrid Topology

Hybrid topology is a network design that combines two or more different types of network topologies into a single integrated structure. This approach allows network designers to take advantage of the strengths of different topologies while minimizing their limitations. Hybrid topology is commonly used in large and complex networks where a single topology type is not sufficient to meet all design requirements.[3]

In hybrid topology, different sections of the network may use different topologies depending on their function. For example, local area networks may be implemented using star topology, while

interconnections between routers may use point-to-point links. This makes hybrid topology highly flexible and scalable, allowing the network to expand without major redesign.

Hybrid topology is widely used in enterprise and campus networks, such as university networks, where multiple buildings and departments are interconnected through a central backbone network. This makes it well suited for the network design implemented in this project.

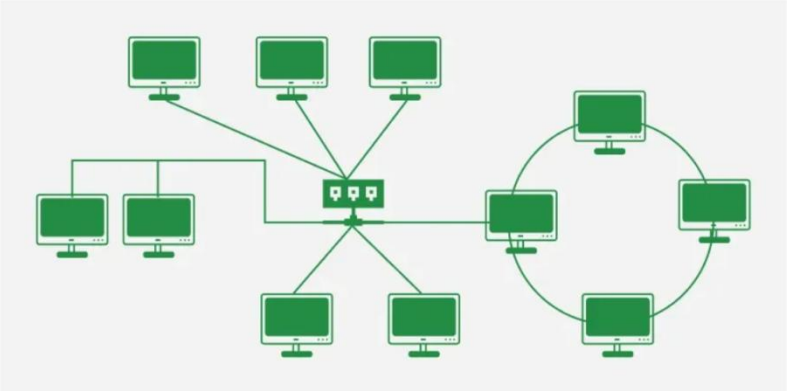


Figure 3:Hybrid Topology[3]

OSPF Routing Protocol

Routing

- Network routing is the process of selecting the best path for data to travel across one or more networks. It plays a vital role in the internet by ensuring that data packets reach their correct destination efficiently.
    - Routing determines the path IP packets follow from source to destination.
    - It is used in packet-switched networks like the Internet.
    - Different routing methods are used to optimize speed, reliability, and performance.[6]

OSPF

Open Shortest Path First (OSPF) is a link-state Interior Gateway Protocol (IGP) developed by the Internet Engineering Task Force (IETF) to provide efficient routing within an Autonomous System (AS). OSPF uses the Shortest Path First (SPF) algorithm, developed by Dijkstra, to determine the best route to each destination.[4]

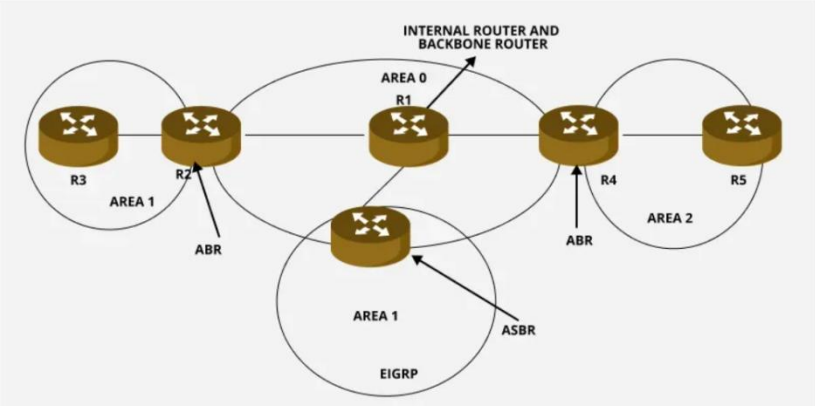


Figure 4:OSPF Network Topology[4]

OSPF operates as a link-state routing protocol. Routers exchange topology information using Link-State Advertisements (LSAs), which are stored in a Link-State Database (LSDB). Each router maintains a complete view of the network topology within its area and applies the Shortest Path First (SPF) algorithm to calculate the most efficient routes.[4]

OSPF networks are divided into areas to improve scalability and reduce routing overhead. The backbone area (Area 0) forms the core of the OSPF network, and all other areas must connect to it. Additional areas, such as normal and stub areas, are used to organize the network and optimize routing efficiency. Routers within OSPF can have different roles, including internal routers, Area Border Routers (ABRs), and Autonomous System Boundary Routers (ASBRs). On broadcast and multi-access networks, OSPF elects a Designated Router (DR) and a Backup Designated Router (BDR) to reduce routing overhead and improve efficiency.[4]

For OSPF routers to form neighbor relationships, several conditions must match, including area ID, router ID, subnet information, timers, and authentication settings. OSPF uses multicast addresses to communicate routing information, where 224.0.0.5 is used for all OSPF routers and 224.0.0.6 is used for DR and BDR communication.[4]

OSPF uses several message types to establish and maintain routing information. Hello messages are used to discover and maintain neighbor relationships, while Database Description, Link-State Request, Link-State Update, and Link-State Acknowledgment messages are used to exchange and synchronize routing databases. Timers such as the Hello timer and Dead timer control neighbor detection and failure recovery.[4]

Overall, OSPF provides fast convergence, loop-free routing, support for load balancing, and efficient route summarization. Despite its relatively higher complexity and resource usage compared to simpler protocols, OSPF is widely used in enterprise and campus networks due to its scalability and reliability, making it suitable for the routing design implemented in this project.

Wireless LAN (WLAN) Concepts

WLAN stands for **Wireless Local Area Network.** WLAN is a local area network that uses radio communication to provide mobility to the network users while maintaining the connectivity to the wired network. A WLAN basically, extends a wired local area network. WLAN's are built by attaching a device called the access point(AP) to the edge of the wired network. Clients communicate with the AP using a wireless network adapter which is similar in function to an ethernet adapter. It is also called a LAWN is a Local area wireless network.

The performance of WLAN is high compared to other wireless networks. The coverage of WLAN is within a campus or building or that tech park. It is used in the mobile propagation of wired networks. The standards of WLAN are HiperLAN, Wi-Fi, and IEEE 802.11. It offers service to the desktop laptop, mobile application, and all the devices that work on the Internet. WLAN is an affordable method and can be set up in 24 hours. WLAN gives users the mobility to move around within a local coverage area and still be connected to the network. Most latest brands are based on IEE 802.11 standards, which are the WI-FI brand name. [5]

Wireless LAN (WLAN) Architecture

Wireless Local Area Network (WLAN) architecture, as defined by IEEE standards, consists of several main components that work together to enable wireless communication. **Stations (STAs)** represent all wireless devices connected to the network, such as laptops and smartphones, and each station is equipped with a wireless network interface controller. A **Basic Service Set (BSS)** is a group of stations that communicate at the physical layer, usually through a single access point. Multiple BSSs can be connected to form an **Extended Service Set (ESS)**, which allows seamless communication and mobility across a wider wireless network. The **Distribution System (DS)** connects all BSSs within an ESS and is responsible for forwarding data between access points, ensuring efficient communication throughout the WLAN.[5]

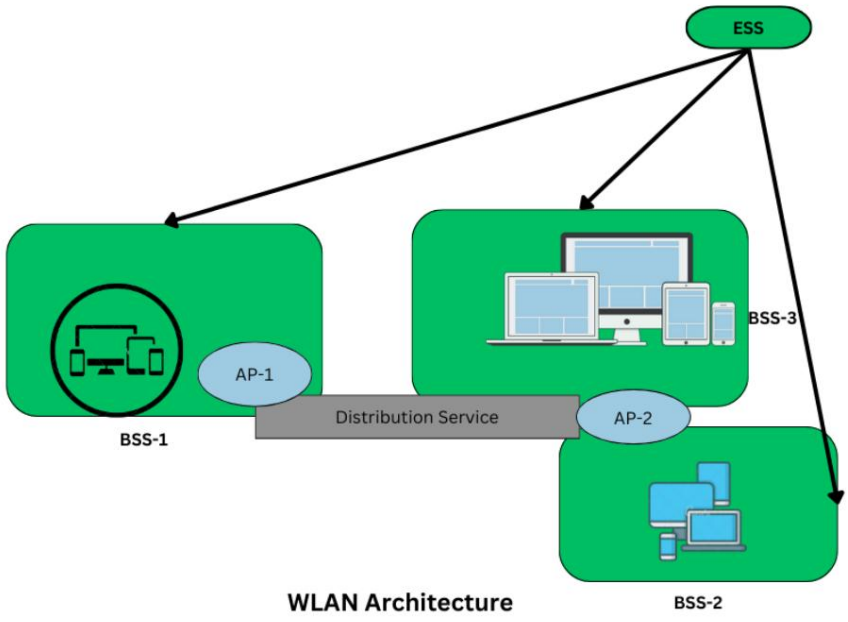


Figure 5:WLAN Architecture[5]

Service Set Identifier (SSID)

The Service Set Identifier (SSID) is the name assigned to a wireless local area network (WLAN). It is used to identify and distinguish one wireless network from another, allowing wireless devices to discover and connect to the correct access point. Each access point broadcasts its SSID so that users can see available wireless networks and select the appropriate one. Using a unique SSID helps organize wireless networks and prevents devices from accidentally connecting to unintended networks, especially in environments where multiple WLANs are present.

Network Devices Used (Conceptual)

Network Devices (or networking hardware) are the physical appliances required for communication and interaction between computers on a computer network.

They act as the "plumbing" of the internet, directing data traffic, amplifying signals, and securing connections. To understand how they work, we must categorize them by the layer of the OSI Model they operate on.[7]

Layer 2 Devices: Data Link Layer

Layer 2 devices operate using MAC addresses and are more intelligent than physical layer devices.

Switch

A switch connects multiple devices within a local area network. Unlike a hub, a switch learns the MAC address of each connected device and forwards data only to the specific port associated with the destination device. This reduces collisions and improves both network performance and security. Switches can be unmanaged, which are plug-and-play devices for small networks, or managed, which support advanced features such as VLANs and Quality of Service (QoS) for enterprise environments.[7]

Access Point (AP)

An access point creates a wireless local area network (WLAN) by allowing wireless devices to connect to a wired Ethernet network. It acts as a bridge between wired and wireless communication. Unlike routers, access points do not perform routing or assign IP addresses unless they are part of a combined wireless router device.[7]

Layer 3 Devices: Network Layer

Layer 3 devices operate using IP addresses and are responsible for connecting different networks.

Router

A router connects multiple networks, such as a local area network to a wide area network (Internet). It uses IP addresses to determine the best path for data packets and maintains a routing table to make forwarding decisions. Routers also block broadcast traffic, which helps isolate networks from each other and improves performance.[7]

Procedure and Discussion:

IP Subnetting and Address Assignment

Main Network Identification

The project started by identifying the main network address provided for the design, which is **102.71.128.0/17**. This address block was used as the base network from which all required subnetworks were derived. The purpose of using a single main network was to ensure unified addressing and organized subnet allocation across the entire topology.

Subnetting Based on Host Requirements

To efficiently divide the main network into multiple subnetworks, Variable Length Subnet Masking (VLSM) was used based on the required number of hosts in each LAN/WLAN. Each segment was assigned the smallest possible subnet mask that can accommodate its host count, which reduces wasted IP addresses and keeps the addressing plan organized. For every subnetwork, the network address, subnet mask, usable host range, and broadcast address were calculated and documented.

Table 1: VLSM Addressing Table

Segment	Hosts Required	Prefix	Subnet Mask	Network Address	Usable IP Address Range	Broadcast Address
C-LAN	500	/23	255.255.254.0	102.71.128.0/23	102.71.128.1 → 102.71.129.254	102.71.129.255
W-LAN1	200	/24	255.255.255.0	102.71.130.0/24	102.71.130.1 → 102.71.130.254	102.71.130.255
W-LAN2	260	/23	255.255.254.0	102.71.132.0/23	102.71.132.1 → 102.71.133.254	102.71.133.255
W-WLAN	480	/23	255.255.254.0	102.71.134.0/23	102.71.134.1 → 102.71.135.254	102.71.135.255
E-LAN1	112	/25	255.255.255.128	102.71.136.0/25	102.71.136.1 → 102.71.136.126	102.71.136.127
E-LAN2	150	/24	255.255.255.0	102.71.137.0/24	102.71.137.1 → 102.71.137.254	102.71.137.255
E-WLAN	550	/22	255.255.252.0	102.71.140.0/22	102.71.140.1 → 102.71.143.254	102.71.143.255

Static IP Assignment for End Devices

After defining the subnet structure, static IP addresses were manually assigned to end devices (PCs) according to the project specifications. Specific host numbers were intentionally selected to verify the correctness of subnetting and IP address calculations. Assigning static IPs ensures stable addressing and reliable inter-network communication between different LANs and WLANs.

Table 2:PC IP Address Assignment Table

Network Segment	Subnet	Assigned PC Host	PC IP Address
C-LAN	102.71.128.0/23	200th host	102.71.128.200
W-LAN1	102.71.130.0/24	50th host	102.71.130.50
W-LAN2	102.71.132.0/23	114th host	102.71.132.114
W-WLAN	102.71.134.0/23	80th host	102.71.134.80
E-LAN1	102.71.136.0/25	30th host	102.71.136.30
E-LAN2	102.71.137.0/24	70th host	102.71.137.70
E-WLAN	102.71.140.0/22	15th host	102.71.140.15

Verification of Addressing Scheme

The assigned IP addresses were verified to ensure that:

- All devices were within the correct subnet range.
- Network and broadcast addresses were not assigned to any device.
- No IP address conflicts existed between devices.

This step confirmed the correctness of the subnetting design before proceeding to router interface configuration and routing setup.

Router Interfaces and Point-to-Point Links Configuration

Router Interconnection Design

After completing the IP addressing for all LANs and WLANs, the next step was to interconnect the routers to enable communication between the Central, West, East, and ISP networks. Point-to-point links were used between routers to provide efficient and clear routing paths.

To minimize IP address waste, all router-to-router links were assigned **/30 subnets**, as each link requires only two usable IP addresses.

Central Network Router Connections

The central router was connected to multiple networks to act as the backbone of the topology. The following point-to-point links were configured:

- **Central ↔ West Core** using the subnet **102.71.144.0/30**
- **Central ↔ C-LAN** using the subnet **102.71.144.4/30**
- **Central ↔ East Core** using the subnet **102.71.144.8/30**
- **Central ↔ ISP** using the subnet **102.71.144.36/30**

Each interface on the central router was assigned the corresponding IP address from its subnet and then enabled to allow packet forwarding.

West Network Router Connections

Within the West network, router interconnections were configured to connect the core router with the West LANs:

- **West Core ↔ W-LAN1** using **102.71.144.12/30**
- **West Core ↔ W-LAN2** using **102.71.144.16/30**
- **W-LAN1 ↔ W-LAN2** internal connection using **102.71.144.32/30**

These links ensured proper communication between West LANs and the rest of the network through the West Core router.

East Network Router Connections

Similarly, router interconnections in the East network were configured as follows:

- **East Core ↔ E-LAN1** using **102.71.144.20/30**
- **East Core ↔ E-LAN2** using **102.71.144.24/30**
- **E-LAN1 ↔ E-LAN2** internal connection using **102.71.144.28/30**

All interfaces were assigned IP addresses according to the subnet plan and activated to allow data transmission.

Table 3:Router Point-to-Point Addressing Table

Link	Subnet (/30)	IP Address (Device 1)	IP Address (Device 2)
Central ↔ West Core	102.71.144.0/30	102.71.144.1	102.71.144.2
Central ↔ C-LAN	102.71.144.4/30	102.71.144.5	102.71.144.6
Central ↔ East Core	102.71.144.8/30	102.71.144.9	102.71.144.10
West Core ↔ W-LAN1	102.71.144.12/30	102.71.144.13	102.71.144.14
West Core ↔ W-LAN2	102.71.144.16/30	102.71.144.17	102.71.144.18
East Core ↔ E-LAN1	102.71.144.20/30	102.71.144.21	102.71.144.22
East Core ↔ E-LAN2	102.71.144.24/30	102.71.144.25	102.71.144.26
E-LAN1 ↔ E-LAN2	102.71.144.28/30	102.71.144.29	102.71.144.30
W-LAN1 ↔ W-LAN2	102.71.144.32/30	102.71.144.33	102.71.144.34
ISP ↔ Central	102.71.144.36/30	102.71.144.37	102.71.144.38

Interface Activation and Verification

After assigning IP addresses, all router interfaces were enabled to ensure operational connectivity. Interface status was checked to confirm that links were up and running. This step ensured that the physical and logical router connections were correctly established before configuring the routing protocol.

OSPF Routing Configuration

Enabling OSPF on Routers

After completing router interconnections, the Open Shortest Path First (OSPF) routing protocol was configured on all routers to enable dynamic routing across the entire network. A single OSPF process ID was used on all routers to maintain consistency. Any previous OSPF configuration was removed before applying the new configuration to avoid conflicts.

OSPF Configuration in the West Network (Area 20)

Routers in the West network were configured as part of **Area 20**. Each router advertised its directly connected LANs and point-to-point links to OSPF using appropriate network statements and wildcard masks.

- **W-LAN1 router** advertised:
    - The W-LAN1 subnet (**102.71.130.0/24**)
    - The link to the West Core router (**102.71.144.12/30**)
    - The internal West router link (**102.71.144.32/30**)
- **W-LAN2 router** advertised:
    - The W-LAN2 subnet (**102.71.132.0/23**)
    - The link to the West Core router (**102.71.144.16/30**)
    - The internal West router link (**102.71.144.32/30**)

The **West Core router** connected Area 20 to the backbone network and therefore advertised:

- The West WLAN subnet (**102.71.134.0/23**) in Area 20
- The links to W-LAN1 and W-LAN2
- The point-to-point link connecting to the Central network as part of **Area 0**

OSPF Configuration in the Central Network (Area 0)

The Central network was configured as the OSPF backbone (**Area 0**). The Central router advertised all point-to-point links connecting it to other networks, including the links to the West Core, East Core, C-LAN, and the ISP router.

The **C-LAN router** was also configured within Area 0 and advertised:

- The C-LAN subnet (**102.71.128.0/23**)
- The point-to-point link connecting it to the Central router

This configuration ensured that all non-backbone areas could communicate through the backbone.

OSPF Configuration in the East Network (Area 30)

Routers in the East network were configured as part of **Area 30**. Each router advertised its connected LANs and router links.

- **E-LAN1 router** advertised:
  - The E-LAN1 subnet (**102.71.136.0/25**)
  - The links to the East Core router and internal East router (**102.71.144.20/30** and **102.71.144.28/30**)
- **E-LAN2 router** advertised:
  - The E-LAN2 subnet (**102.71.137.0/24**)
  - The links to the East Core router and internal East router (**102.71.144.24/30** and **102.71.144.28/30**)

The **East Core router** advertised:

- The East WLAN subnet (**102.71.140.0/22**) in Area 30
- The links to E-LAN1 and E-LAN2
- The point-to-point link to the Central router in **Area 0**

ISP Router OSPF Configuration

The ISP router was configured to participate in OSPF within **Area 0**. The point-to-point link between the ISP and the Central router was advertised, ensuring full routing visibility and end-to-end connectivity across the network.

OSPF Verification

After completing the OSPF configuration, routing tables were checked on all routers to verify that routes from all areas were successfully learned. Neighbor relationships were confirmed to ensure that OSPF adjacencies were established correctly between connected routers.

Results and Discussion

This part explains how we designed the network, how we configured it, and how we tested that everything works. We built the network in Cisco Packet Tracer using three areas (Area 0, Area 20, and Area 30). We used static IP addresses based on the leader Student ID (**0271**), and we created the required subnets so each LAN and WLAN has enough IP addresses for the number of devices it must support.

4.1 IP addressing plan and topology labels

The base network (using the Student ID) is **102.71.128.0/17**. We divided this network into smaller subnets for each LAN and WLAN, and we used small /30 subnets for the router-to-router links. In the Packet Tracer file, we added clear labels for every network, router, and link so the topology is easy to understand.

Table 4 : Subnet Allocation for LANs and WLANs

Network name	Network/CIDR	Subnet mask	Default gateway	Required hosts
C-LAN	102.71.128.0/23	255.255.254.0	102.71.128.1	500
W-LAN1	102.71.130.0/24	255.255.255.0	102.71.130.1	200
W-LAN2	102.71.132.0/23	255.255.254.0	102.71.132.1	260
W-WLAN	102.71.134.0/23	255.255.254.0	102.71.134.1	480
E-LAN1	102.71.136.0/25	255.255.255.128	102.71.136.1	112
E-LAN2	102.71.137.0/24	255.255.255.0	102.71.137.1	150
E-WLAN	102.71.140.0/22	255.255.252.0	102.71.140.1	550

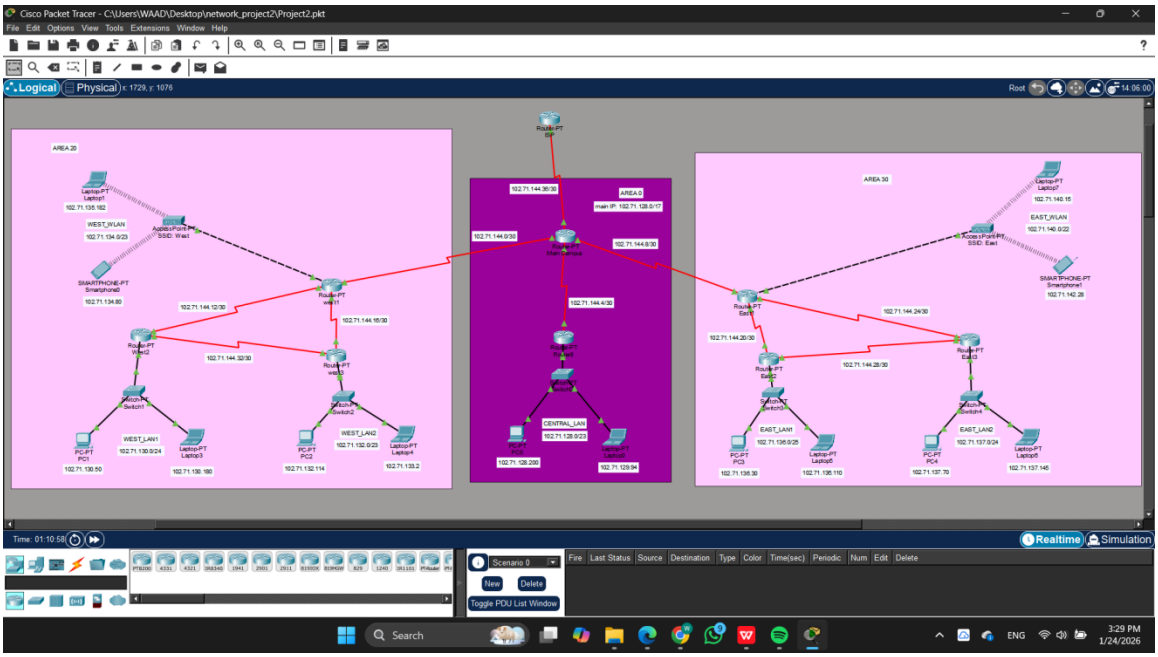


Figure 6 : Labeled topology in Packet Tracer

This figure shows the full network in Packet Tracer with clear labels for the areas, routers, LAN/WLAN names, and the router-to-router link networks.

4.2 Static IP configuration for routers

Each router interface was given a fixed IPv4 address and subnet mask based on the addressing plan. The links between routers use **/30 networks** because they only need two usable IP addresses, so this saves IP space. This part shows proof that the IP settings are correct on all routers.

Table 5 : Serial Links IP Addressing Summary

Link	/30 Network	Router A IP	Router B IP
Main Campus ↔ West1	102.71.144.0/30	102.71.144.2	102.71.144.1
Main Campus ↔ Router 8	102.71.144.4/30	102.71.144.5	102.71.144.6
Main Campus ↔ East1	102.71.144.8/30	102.71.144.9	102.71.144.10
West1 ↔ West2	102.71.144.12/30	102.71.144.13	102.71.144.14
West1 ↔ West3	102.71.144.16/30	102.71.144.17	102.71.144.18
West2 ↔ West3	102.71.144.32/30	102.71.144.33	102.71.144.34
East1 ↔ East2	102.71.144.20/30	102.71.144.21	102.71.144.22
East1↔ East3	102.71.144.24/30	102.71.144.25	102.71.144.26
East2 ↔ East3	102.71.144.28/30	102.71.144.29	102.71.144.30
ISP Router ↔ Main Campus	102.71.144.36/30	102.71.144.37	102.71.144.38

This table shows the IP addresses used on the **router-to-router links**. Each link uses a **/30 subnet** because it gives **two usable IP addresses**, one for each router on that link, which saves IP space. The table lists the subnet for each connection and the IP address on both ends. These settings were checked later using show ip interface brief and show ip route.

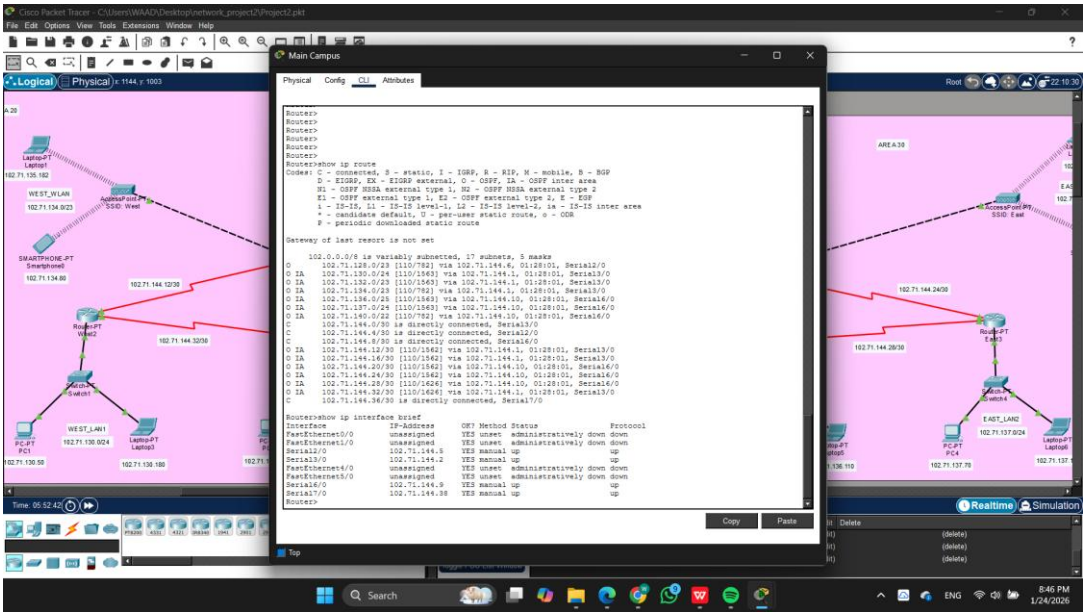


Figure 7:Router static IP configuration (Main Campus)

This screenshot shows show ip interface brief confirming the Main Campus serial interfaces have the planned /30 IPs: **S3/0=102.71.144.2**, **S6/0=102.71.144.9**, **S2/0=102.71.144.5**, **S7/0=102.71.144.38**.

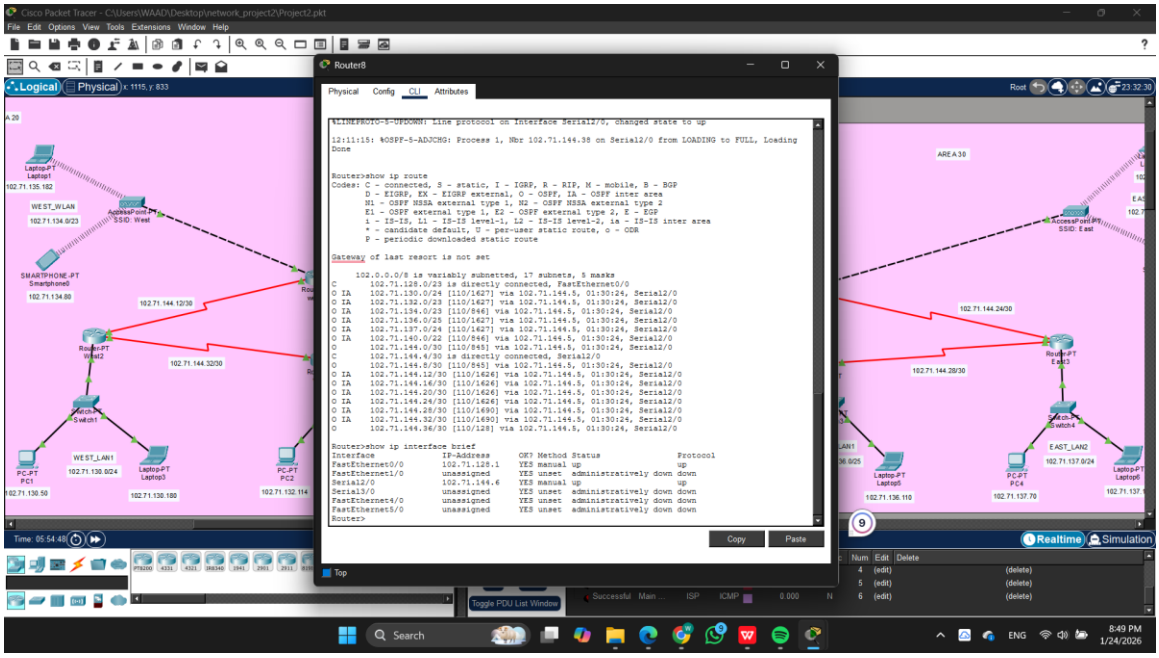


Figure 8 :Router static IP configuration (Router8)

This screenshot shows the LAN interface IP for the Central LAN and the link back to Main Campus: **Fa0/0 = 102.71.128.1/23** and **Serial2/0 = 102.71.144.6/30**.

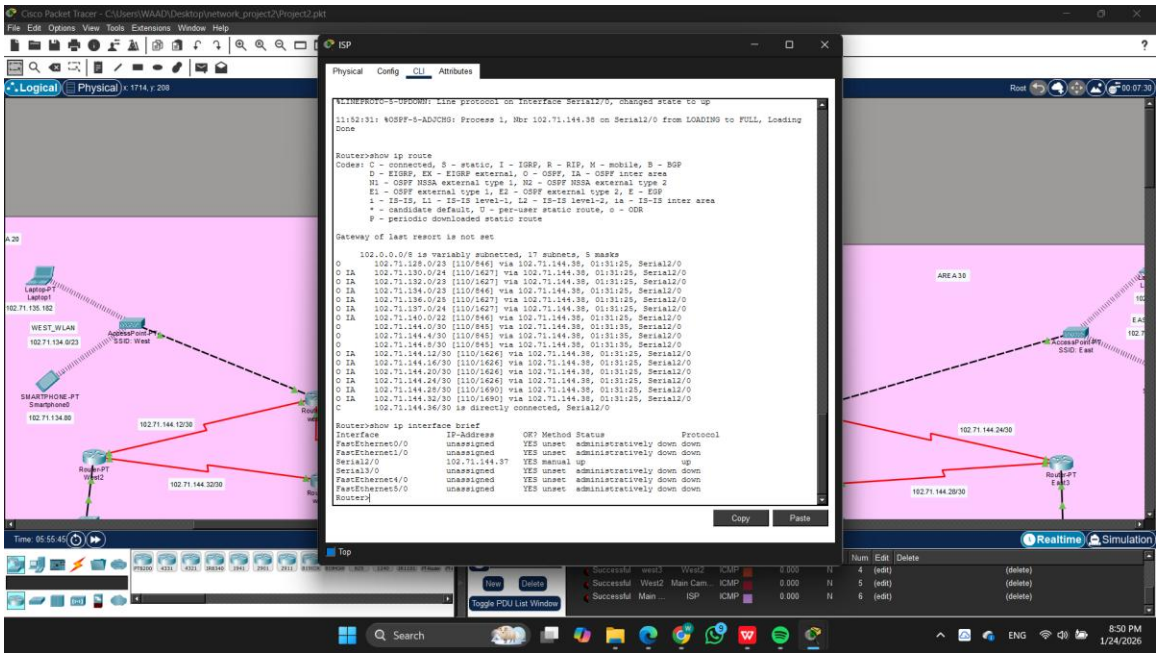


Figure 9 : Router static IP configuration (ISP)

This screenshot confirms the ISP router has the correct point-to-point IP on the link to Main Campus: **Serial2/0 = 102.71.144.37/30**.

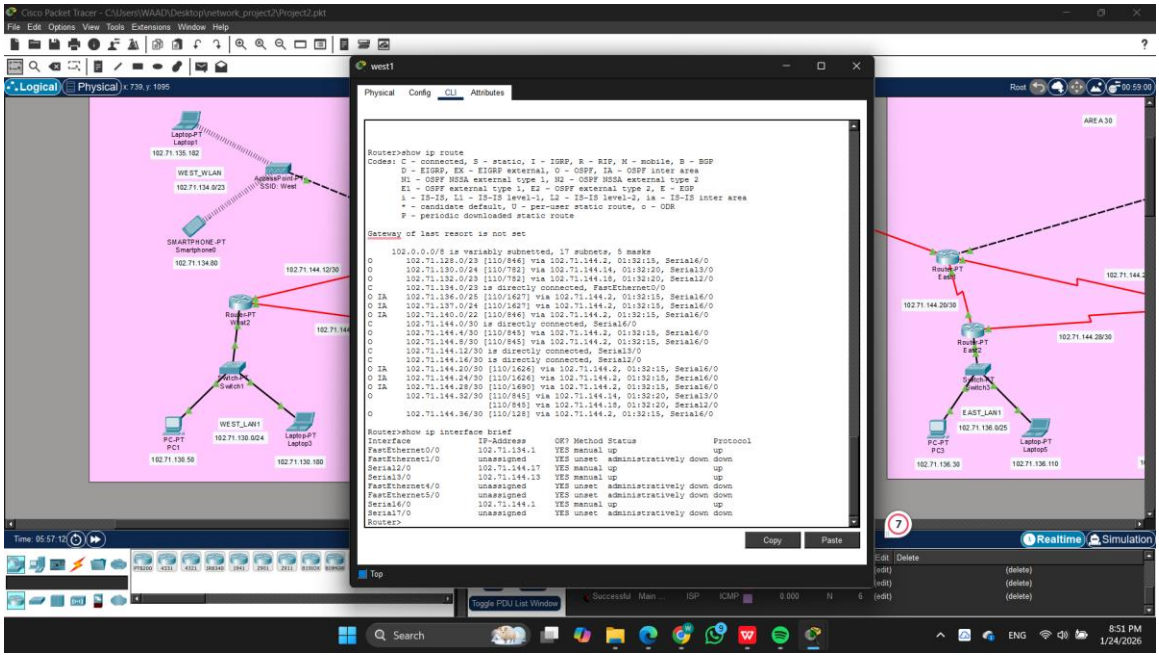


Figure 10 : Router static IP configuration (West1)

This screenshot confirms West1 has the correct LAN and serial IPs: **Fa0/0 = 102.71.134.1/23**, **Serial6/0 = 102.71.144.1/30**, **Serial3/0 = 102.71.144.13/30**, **Serial2/0 = 102.71.144.17/30**.

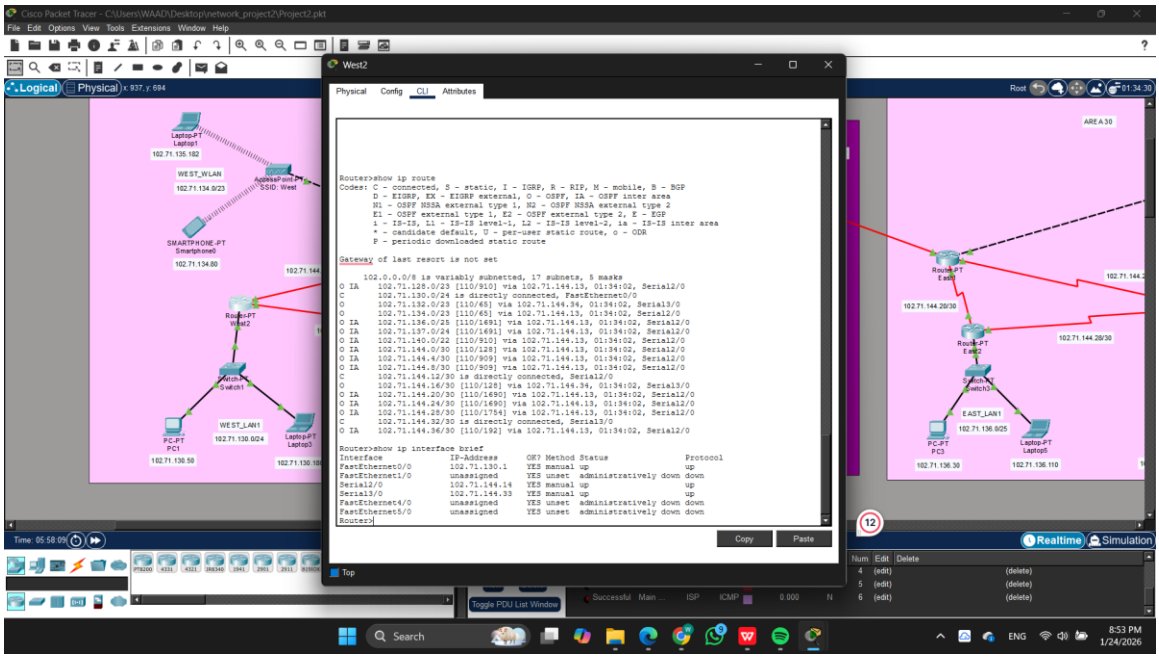


Figure 11 : Router static IP configuration (West2)

This screenshot shows West2 addressing for the W-LAN1 network and its serial links: **Fa0/0 = 102.71.130.1/24**, **Serial2/0 = 102.71.144.14/30**, **Serial3/0 = 102.71.144.33/30**.

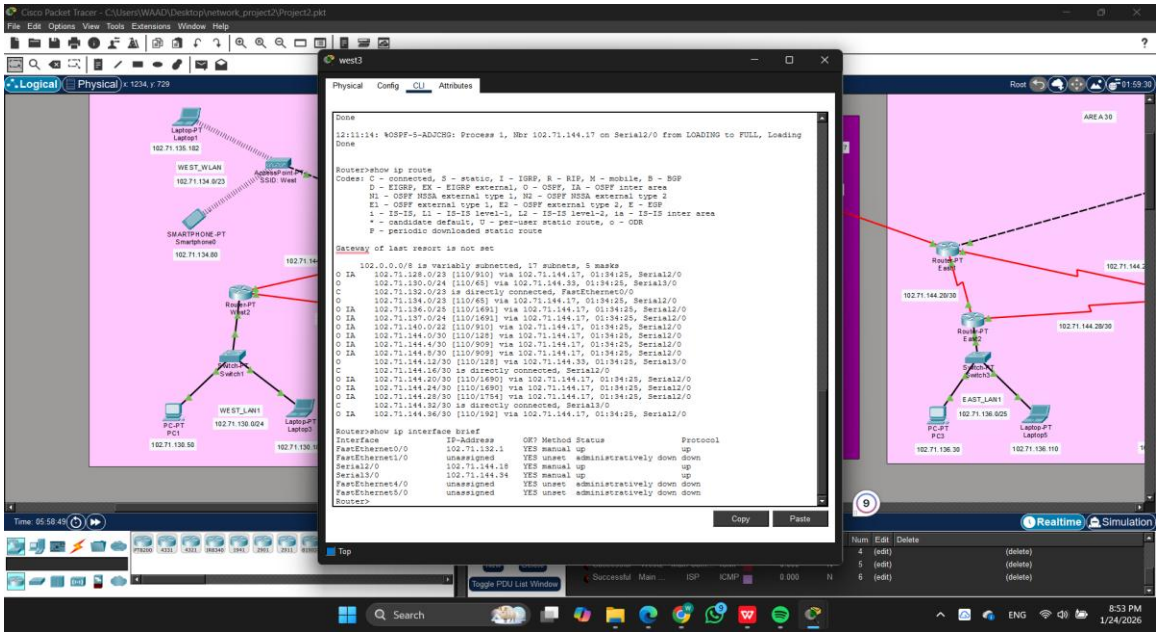


Figure 12 : Router static IP configuration (West3)

This screenshot confirms West3 IP settings: **Fa0/0 = 102.71.132.1/23**, **Serial2/0 = 102.71.144.18/30**, **Serial3/0 = 102.71.144.34/30**.

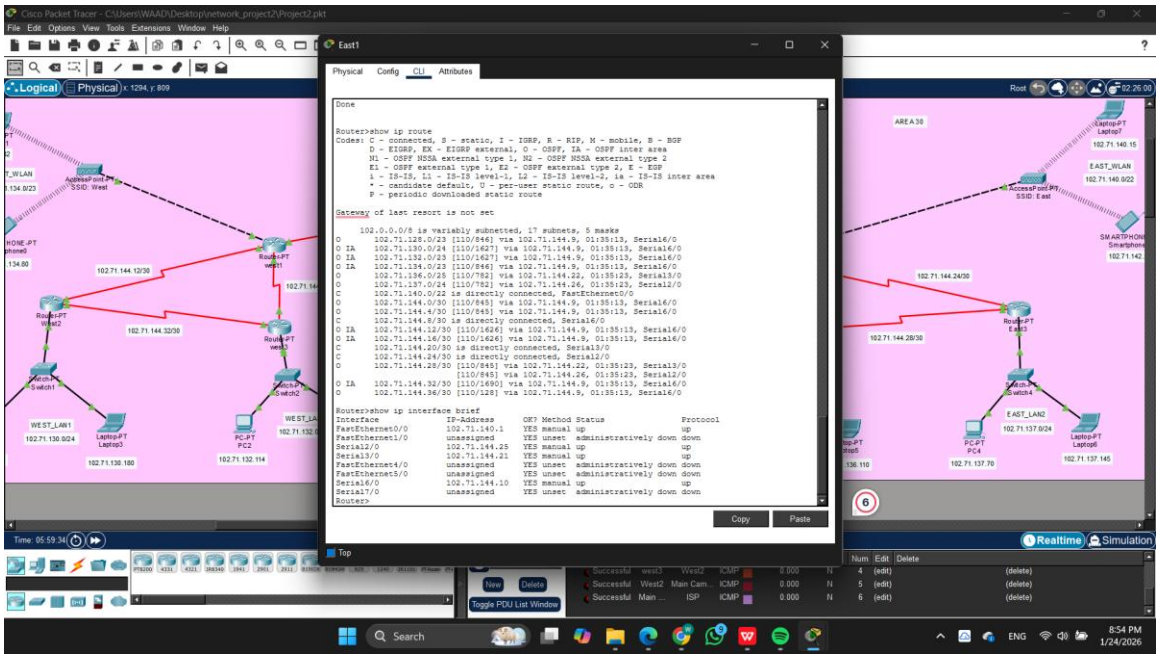


Figure 13 : Router static IP configuration (East1)

This screenshot shows East1 has the correct LAN and serial addressing: **Fa0/0 = 102.71.140.1/22**, **Serial6/0 = 102.71.144.10/30**, **Serial3/0 = 102.71.144.21/30**, **Serial2/0 = 102.71.144.25/30**.

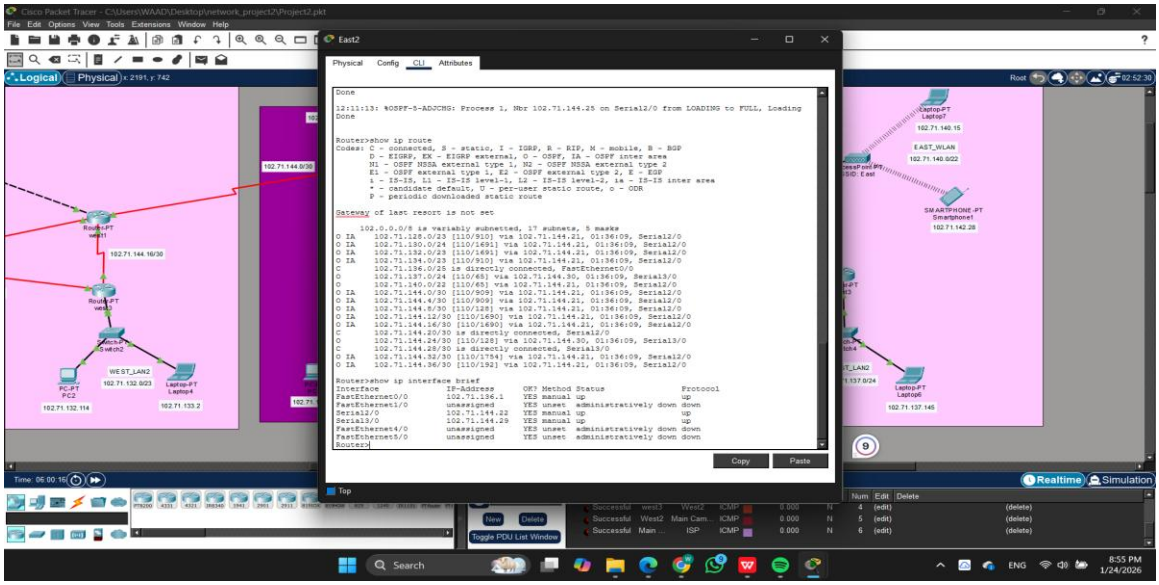


Figure 14 : Router static IP configuration (East2)

This screenshot confirms East2 addressing: **Fa0/0 = 102.71.136.1/25**, **Serial2/0 = 102.71.144.22/30**, **Serial3/0 = 102.71.144.29/30**.

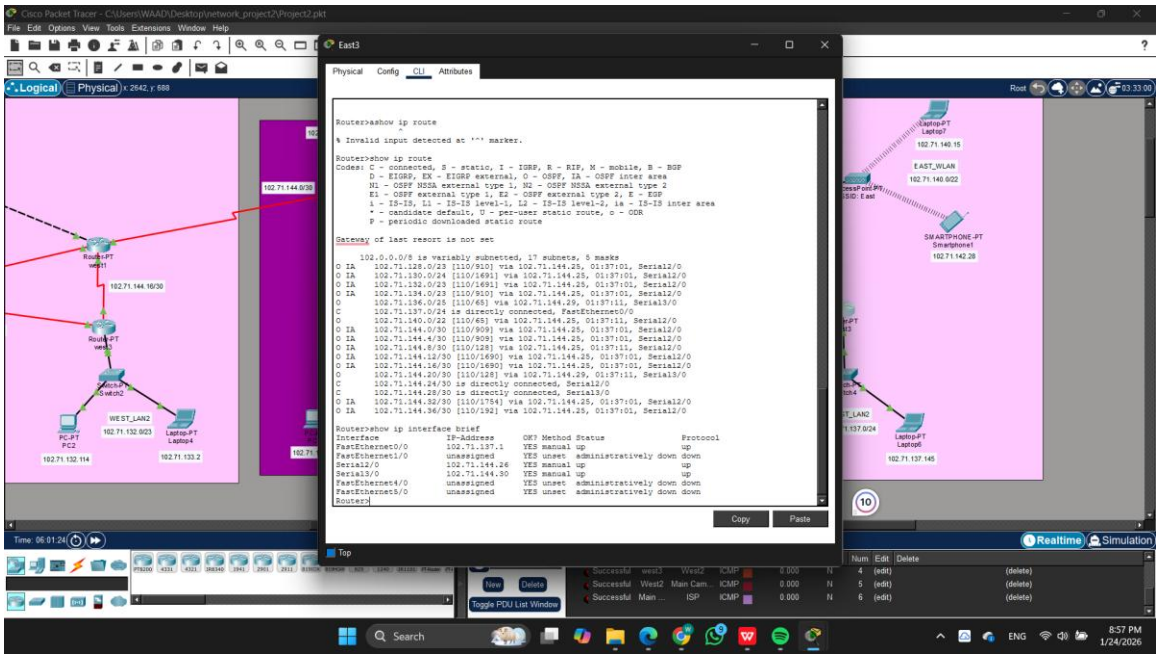


Figure 15 : Router static IP configuration (East3)

This screenshot shows East3 has correct IPs: **Fa0/0 = 102.71.137.1/24**, **Serial2/0 = 102.71.144.26/30**, **Serial3/0 = 102.71.144.30/30**.

Discussion:

The screenshots show that each router has the correct IP address on its interfaces based on the addressing plan. Having the correct static IP settings is important because OSPF can only form neighbor connections when the two routers on the same link are in the **same subnet**. If there is a mistake like a wrong subnet mask, a wrong IP on a link, or two devices using the same IP, OSPF neighbors may not connect and routing between the areas will fail.

IP configuration for end devices

End devices were configured with static IPv4 addresses that satisfy the task's 'Nth host' requirement within each subnet. Default gateways were set to the corresponding router LAN interface.

Table 6 : End-device addressing

Device	Network name	IP address	Mask	Default gateway	
PC0	C-LAN	102.71.128.200	255.255.254.0	102.71.128.1	200th host
Laptop0	C-LAN	102.71.129.94	255.255.254.0	102.71.128.1	350th host
PC1	W-LAN1	102.71.130.50	255.255.255.0	102.71.130.1	50th host
Laptop3	W-LAN1	102.71.130.180	255.255.255.0	102.71.130.1	180th host
PC2	W-LAN2	102.71.132.114	255.255.254.0	102.71.132.1	114th host
Laptop4	W-LAN2	102.71.133.2	255.255.254.0	102.71.132.1	258th host
Smartphone0	W-WLAN	102.71.134.80	255.255.254.0	102.71.134.1	80thhost
Laptop1	W-WLAN	102.71.135.182	255.255.254.0	102.71.134.1	438th host
PC3	E-LAN1	102.71.136.30	255.255.255.128	102.71.136.1	30th host
Laptop5	E-LAN1	102.71.136.110	255.255.255.128	102.71.136.1	110th host
PC4	E-LAN2	102.71.137.70	255.255.255.0	102.71.137.1	70th host
Laptop6	E-LAN2	102.71.137.145	255.255.255.0	102.71.137.1	145th host
Smartphone1	E-WLAN	102.71.140.15	255.255.252.0	102.71.140.1	15th host
Laptop2	E-WLAN	102.71.142.28	255.255.252.0	102.71.140.1	540th host

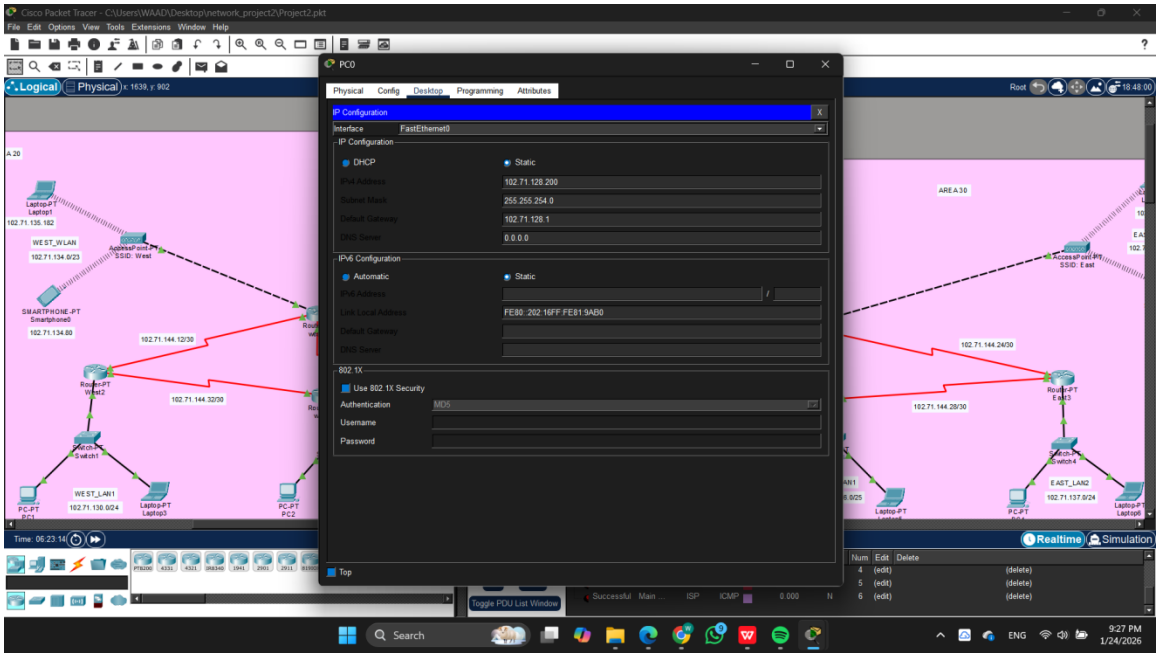


Figure 16 : Central LAN end device IP configuration (PC0)

This screenshot shows PC0 using a **static IPv4 address** in the Central LAN network. Important values: **IP = 102.71.128.200**, **Mask = 255.255.254.0**, **Default Gateway = 102.71.128.1**.

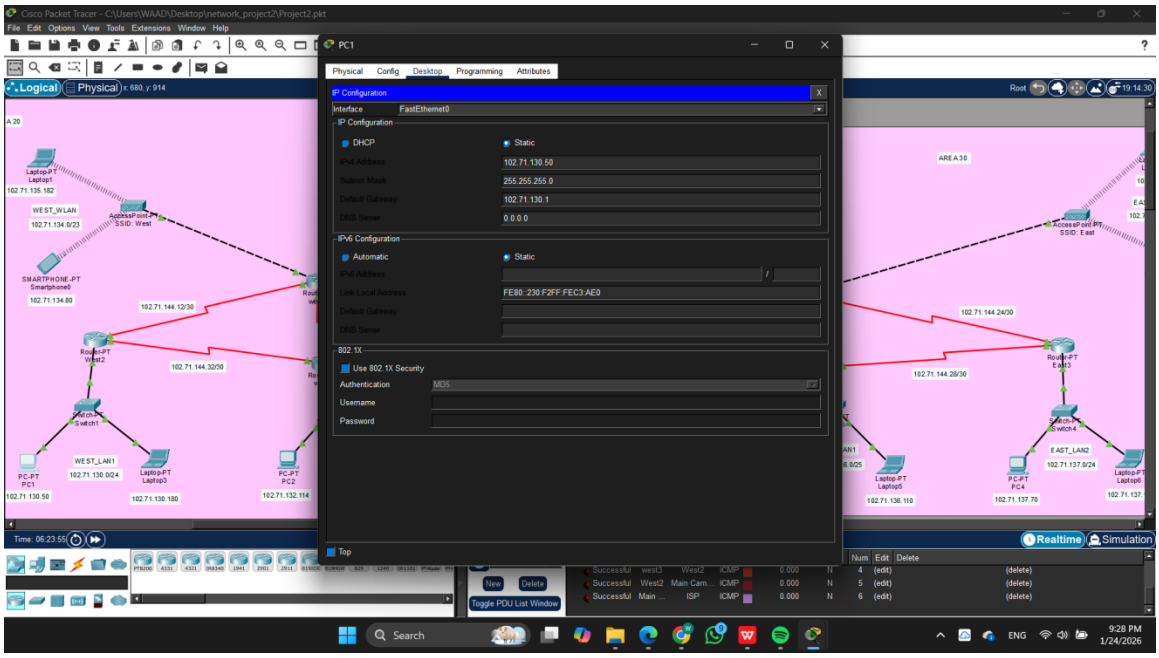


Figure 17 : West LAN1 end device IP configuration (PC1)

This screenshot shows PC1 configured with a static IP in the West LAN1 subnet. Important values: **IP = 102.71.130.50**, **Mask = 255.255.255.0**, **Default Gateway = 102.71.130.1**.

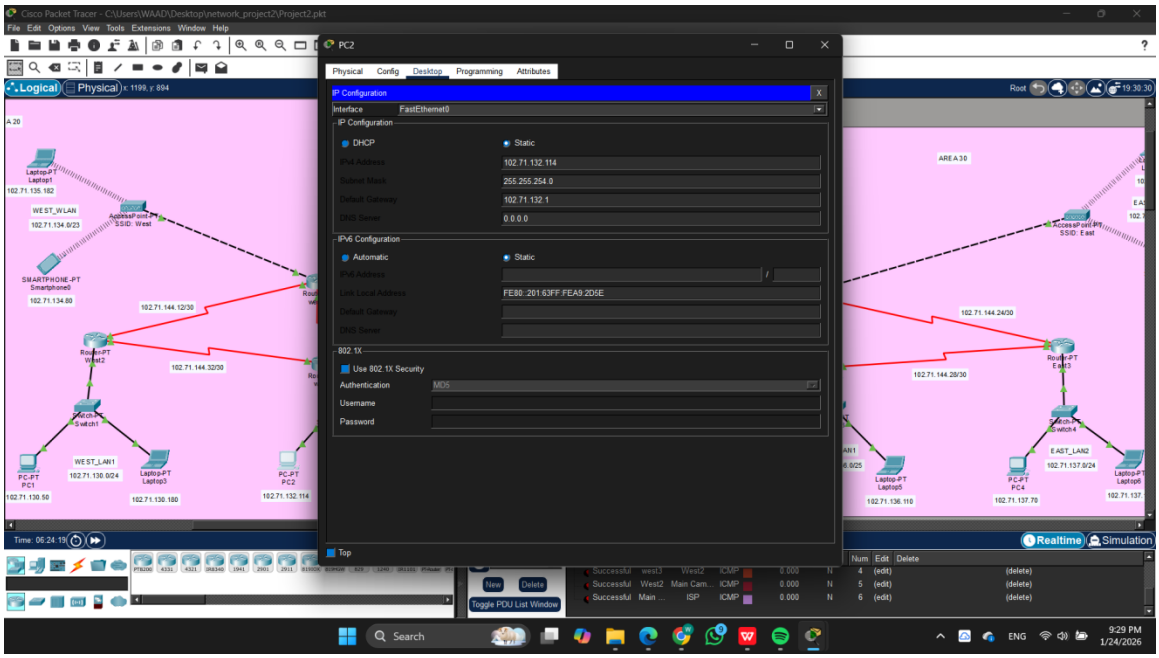


Figure 18 : West LAN2 end device IP configuration (PC2)

This screenshot shows PC2 using a static IP address in the West LAN2 network.

Important values: **IP = 102.71.132.114**, **Mask = 255.255.254.0**, **Default Gateway = 102.71.132.1**.

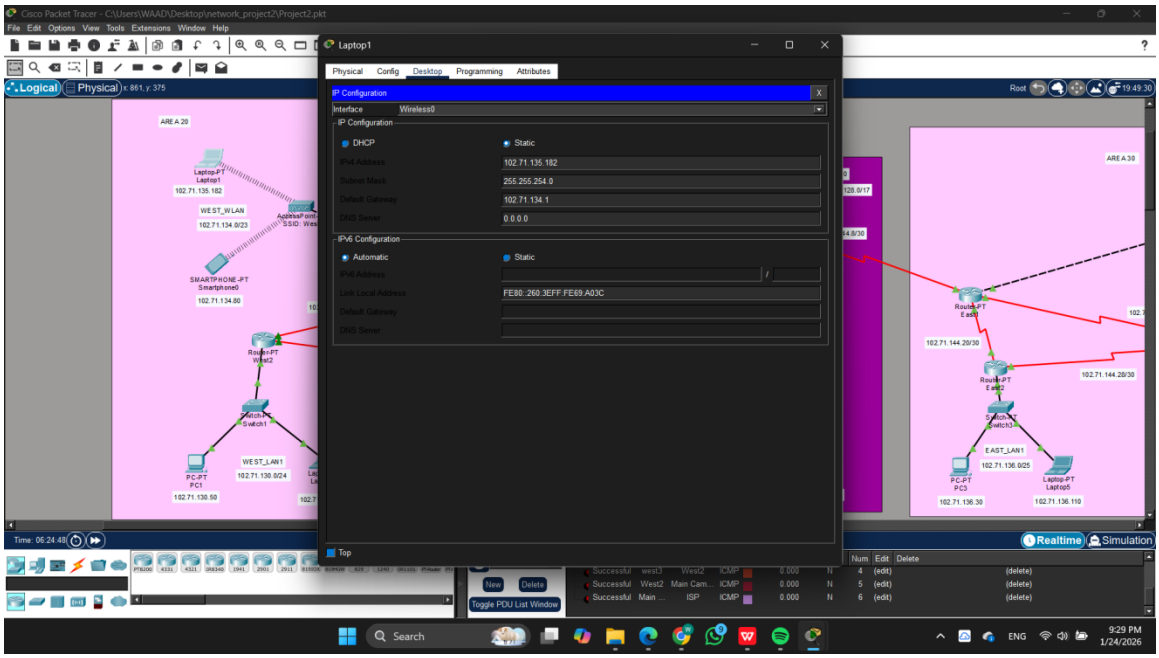


Figure 19 : West WLAN device IP configuration (Laptop1)

This screenshot shows Laptop1 connected on the West wireless network with a static IP.

Important values: **IP = 102.71.135.182**, **Mask = 255.255.254.0**, **Default Gateway = 102.71.134.1**.

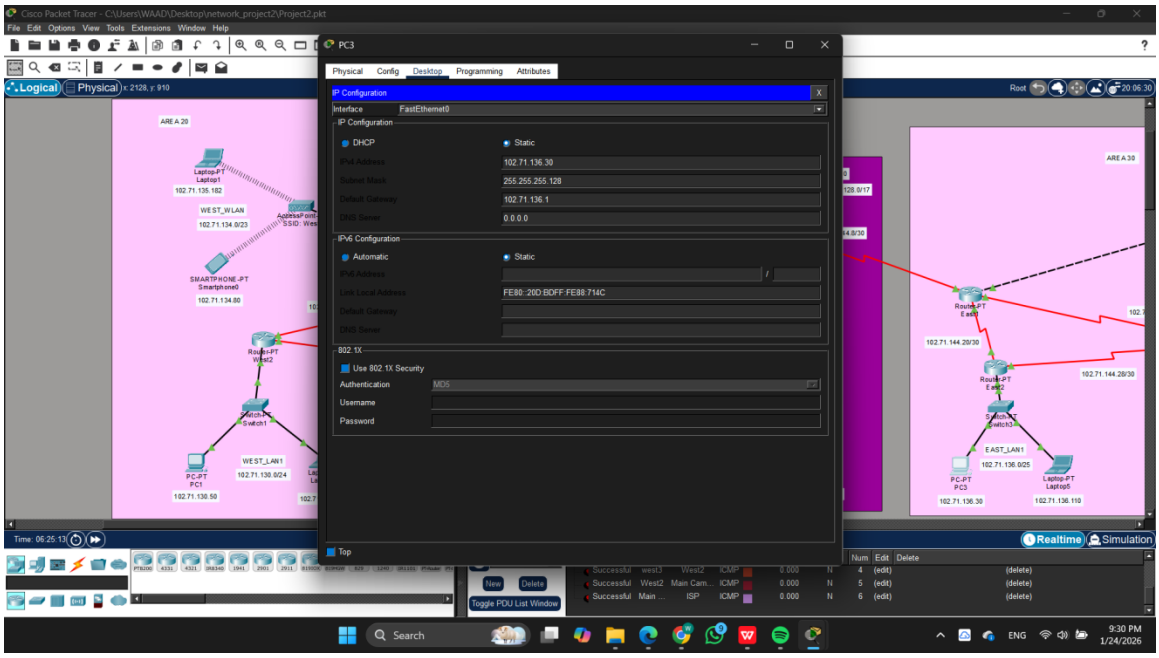


Figure 20 : East LAN1 end device IP configuration (PC3)

This screenshot shows PC3 configured in the East LAN1 subnet using static addressing. Important values: **IP = 102.71.136.30**, **Mask = 255.255.255.128**, **Default Gateway = 102.71.136.1**.

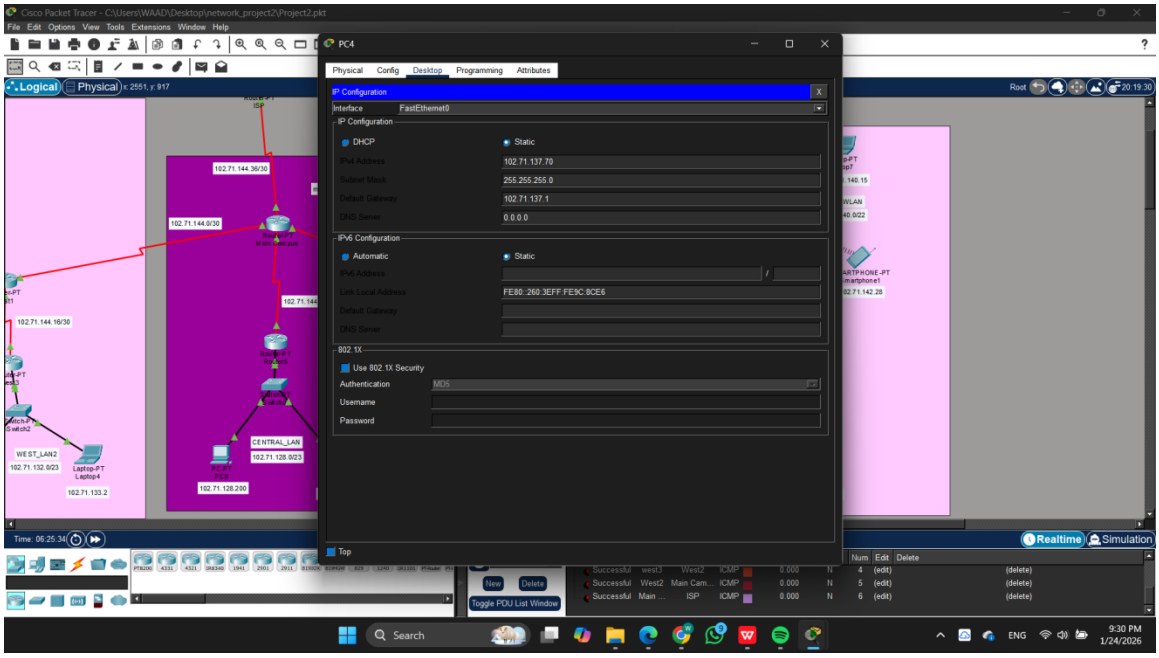


Figure 21 : East LAN2 end device IP configuration (PC4)

This screenshot shows PC4 using a static IP address in the East LAN2 network. Important values: **IP = 102.71.137.70**, **Mask = 255.255.255.0**, **Default Gateway = 102.71.137.1**.

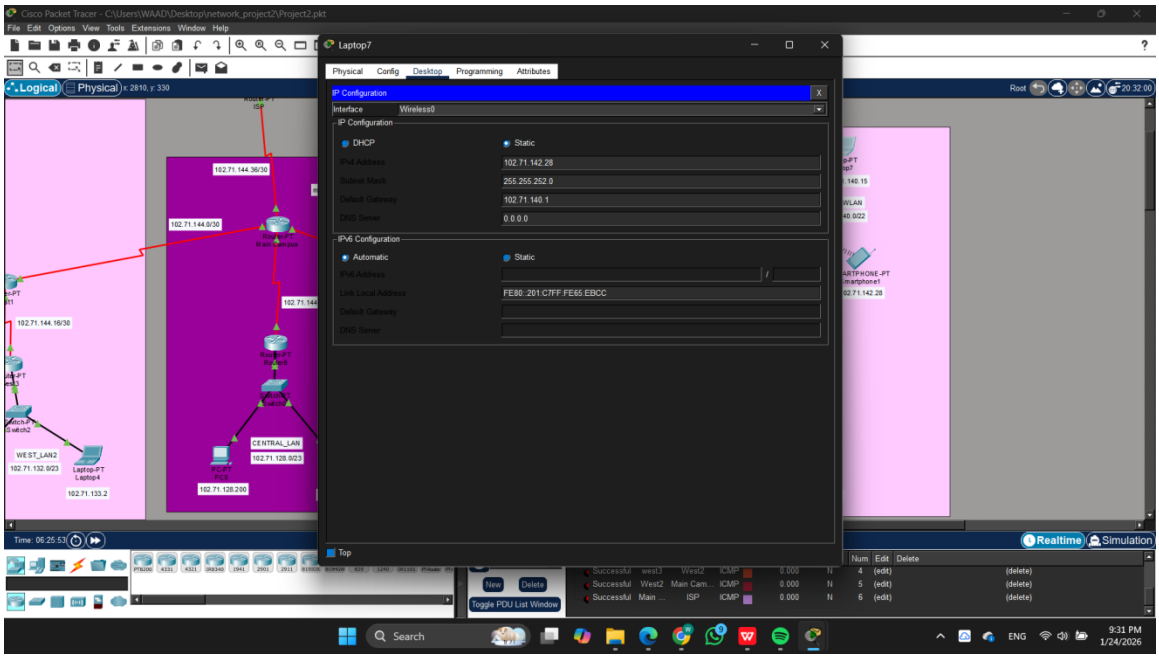


Figure 22 : East WLAN device IP configuration (Laptop7)

This screenshot shows Laptop7 connected on the East wireless network with a static IP. Important values: **IP = 102.71.142.28**, **Mask = 255.255.252.0**, **Default Gateway = 102.71.140.1**.

Discussion:

The screenshots show that each end device has the correct **IP address**, **subnet mask**, and **default gateway** for its network. This is important because the gateway is the router address that the device uses to reach **other networks**. The masks match the subnet sizes in the addressing plan, and the gateway in each device matches the router interface for that subnet. If the IP, mask, or gateway is wrong, the device may work only inside its local network but will fail to ping devices in other areas.

End-to-end connectivity tests (ping and tracert)

Connectivity was validated by testing **ping** and path selection **tracert** between end devices in different areas. Successful ping confirms IP reachability, tracert confirms that traffic follows the expected router path.

1. Same subnet (wired end device to wired end device) LAN(west)

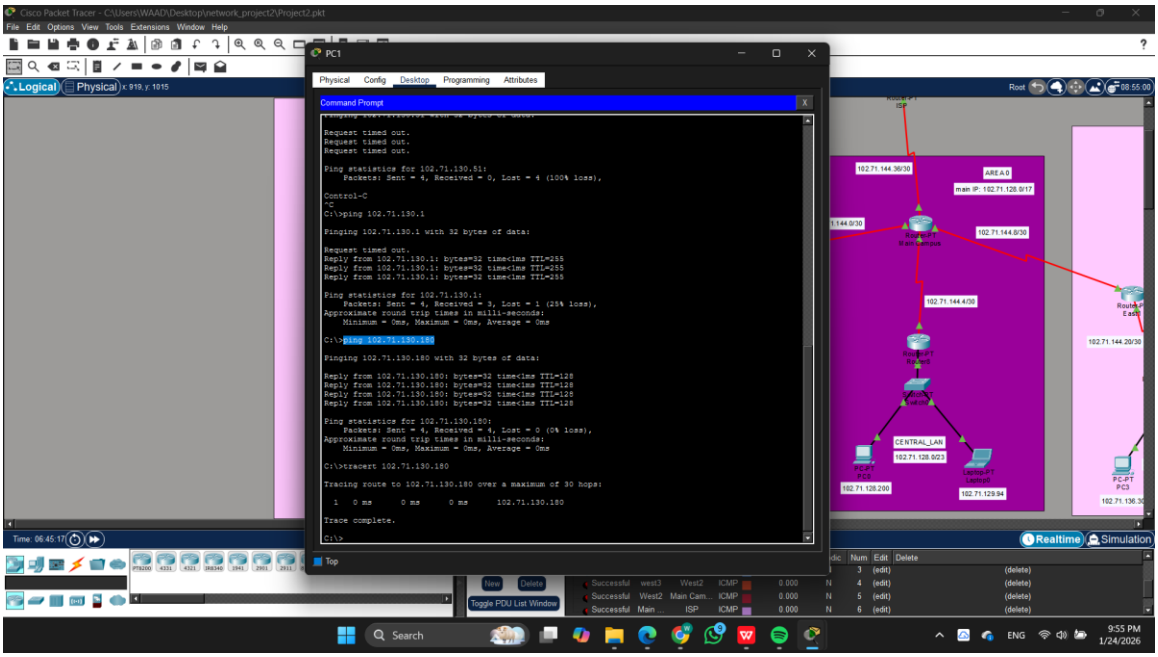


Figure 23 : Same subnet (wired end device to wired end device) LAN(west)

In this picture, **PC1 is testing connectivity to another wired device in the same subnet** using ping. The ping shows **4 replies received and 0 packet loss**. The tracert result also shows the destination reached in **one hop**, which tells us the traffic is staying inside the local network . So **same-subnet wired-to-wired communication is working fine**.

2. Same subnet (wired end device to wired end device) LAN(east)

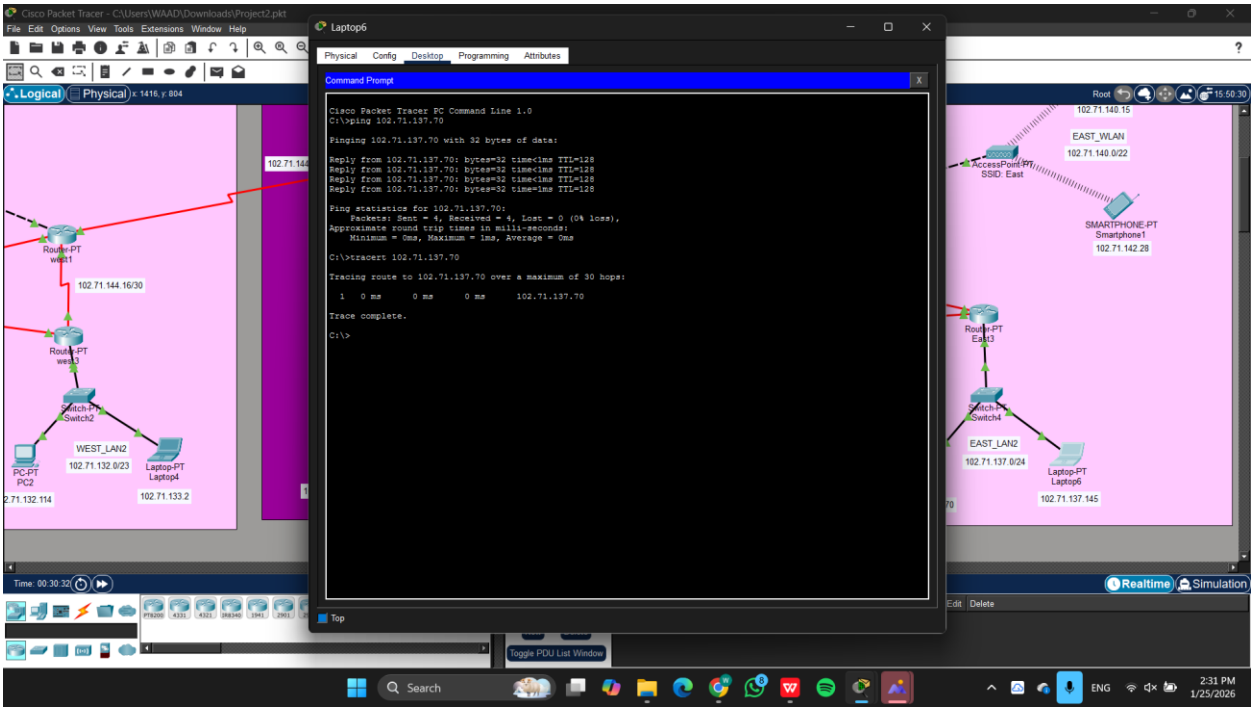


Figure 24 : Same subnet (wired end device to wired end device) LAN(east)

3. Same subnet (wired end device to wired end device) LAN(center)

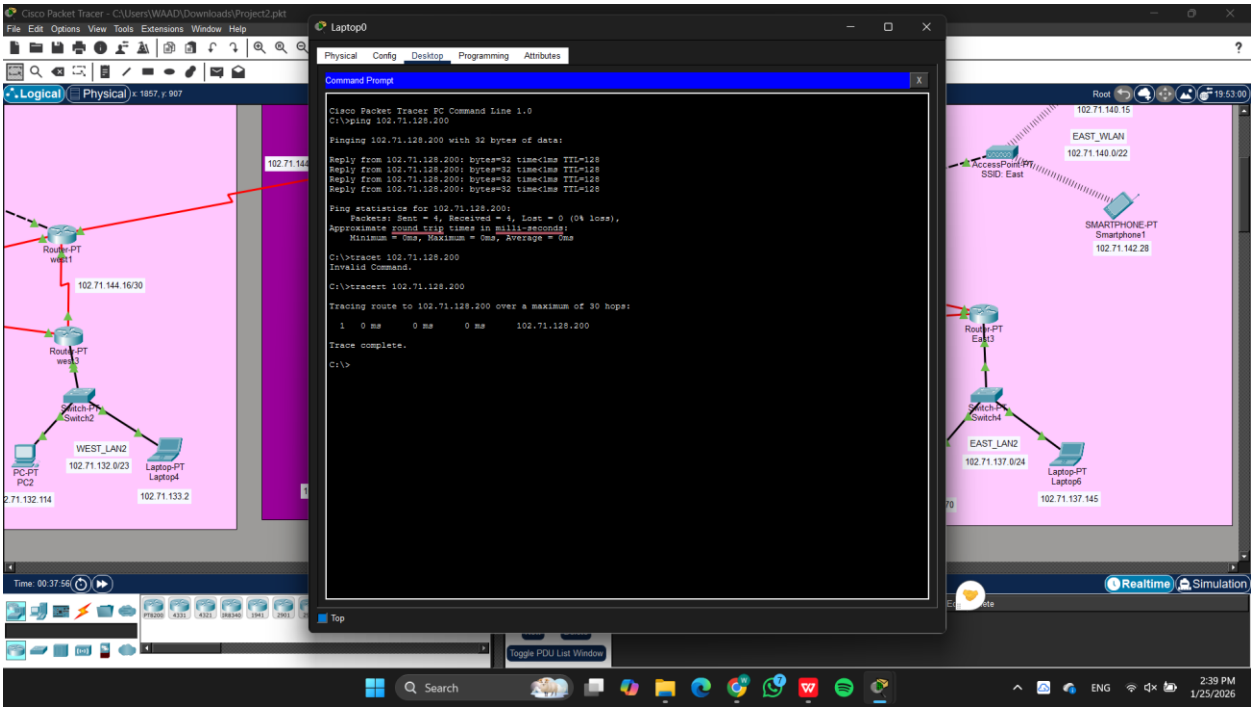


Figure 25 : Same subnet (wired end device to wired end device) LAN(center)

4. Same subnet (wireless end device to wireless end device)

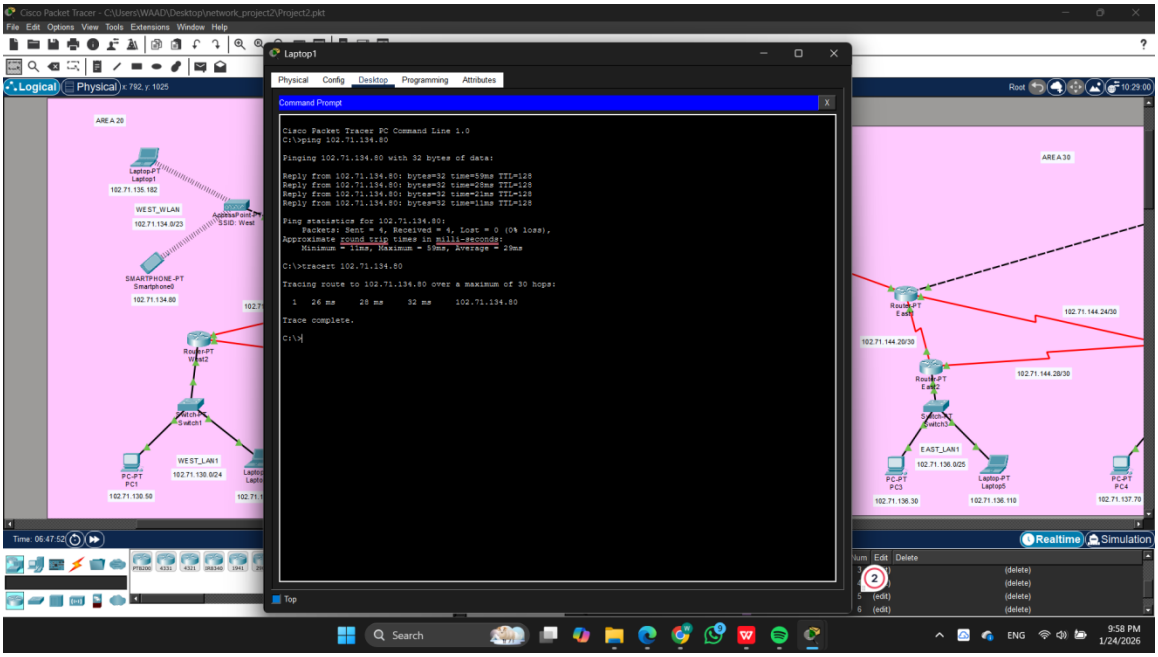


Figure 26: Same subnet (wireless end device to wireless end device)

In this picture, **Laptop1 (a wireless device)** is testing the connection to **another wireless device in the** same network range using ping. The results show **four replies and 0% packet loss**. The tracert also reaches the target in **one hop**, which means the traffic stays inside the local wireless network.

5. Same subnet (wireless end device to wire end device) (west)

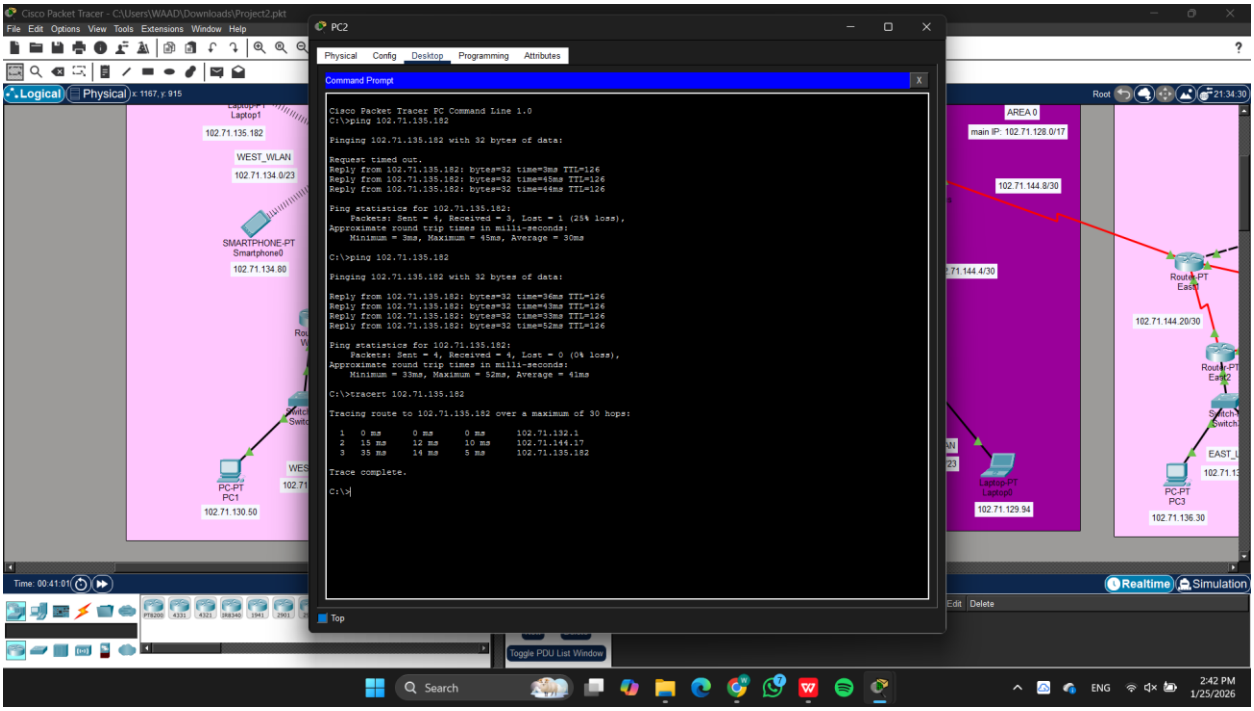


Figure 27 : Same subnet (wireless end device to wire end device) (west)

6. Same subnet (wireless end device to wire end device) (east)

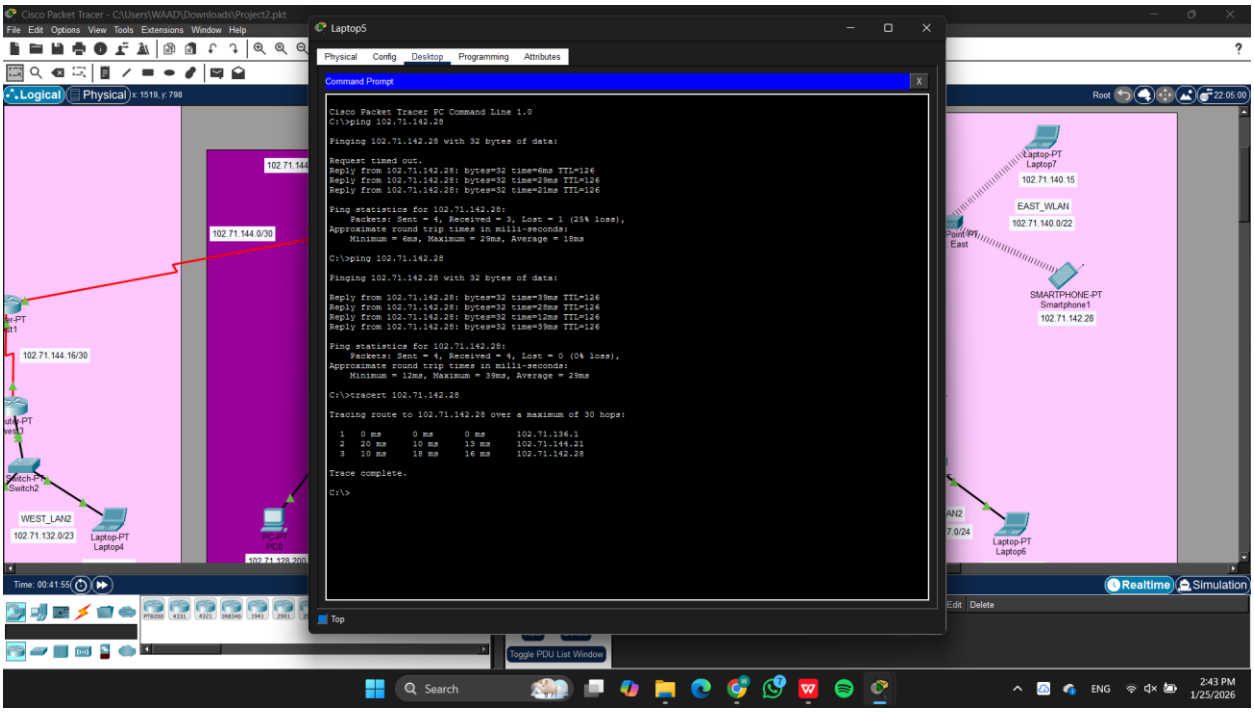


Figure 28 : Same subnet (wireless end device to wire end device) (east)

7. Same area, different subnets (Area 20)

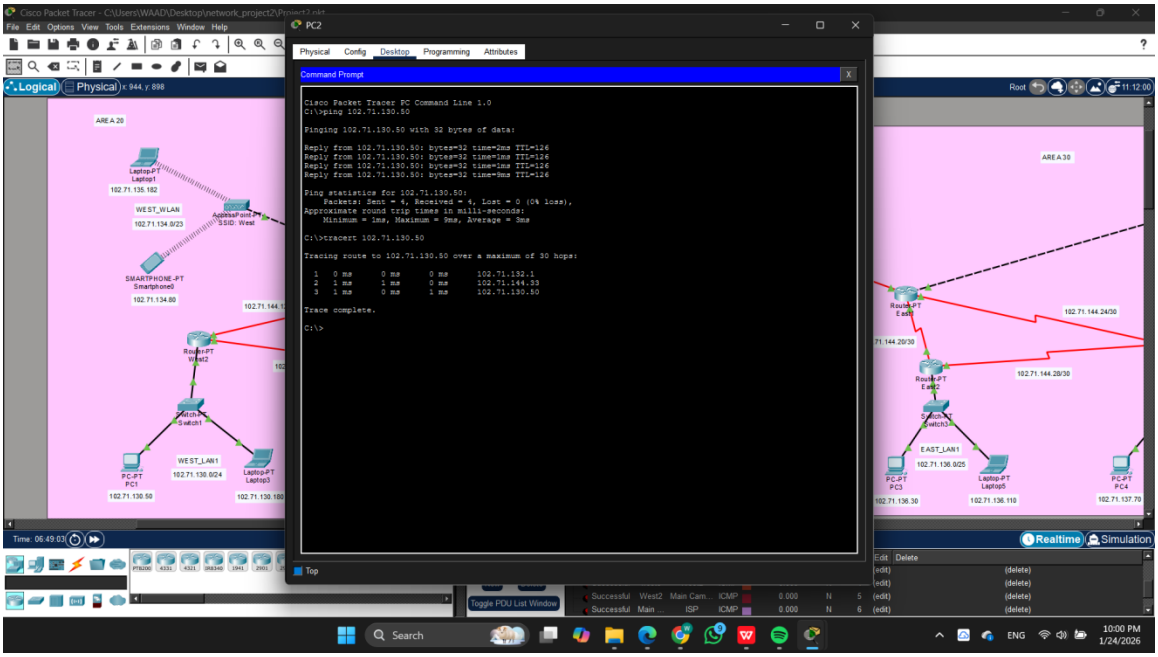


Figure 29 : Same area, different subnets (Area 20)

In this picture, **PC2 is pinging a device that is in a different subnet but still inside the same area (Area 20)**. The ping shows **4 replies and 0% packet loss**. The tracert output also shows the path going **from PC2 to its default gateway (first router), then across another router link, and finally to the destination**, which proves the routers in **Area 20 are sharing routes correctly and forwarding traffic between the subnets.**

8. Same area, different subnets (Area 30)

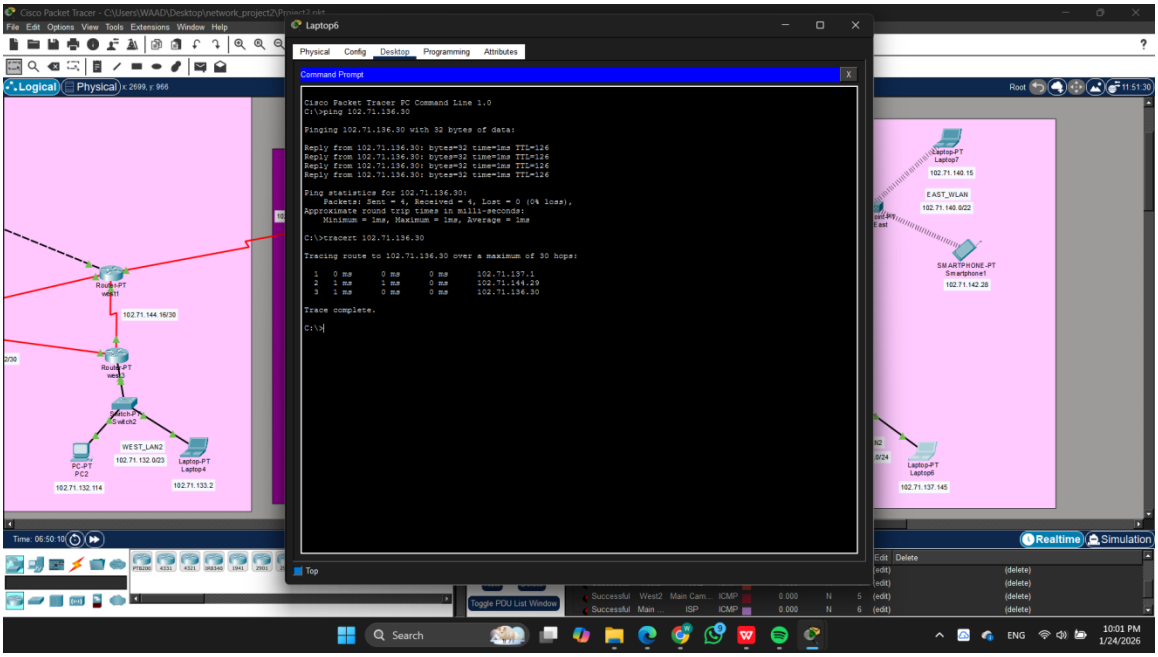


Figure 30 : Same area, different subnets (Area 30)

In this picture, **Laptop6 is testing connectivity to a device in a different subnet but still inside the same area (Area 30)**. The ping results show **4 replies with 0% packet loss**. The tracert output shows the traffic going **from the laptop to its default gateway (first router), then through the routers inside Area 30, and finally to the destination**, proving that **routing within Area 30 is working correctly and the routers are sharing the needed routes.**

9. Inter-area (Area 20 to Area 30)

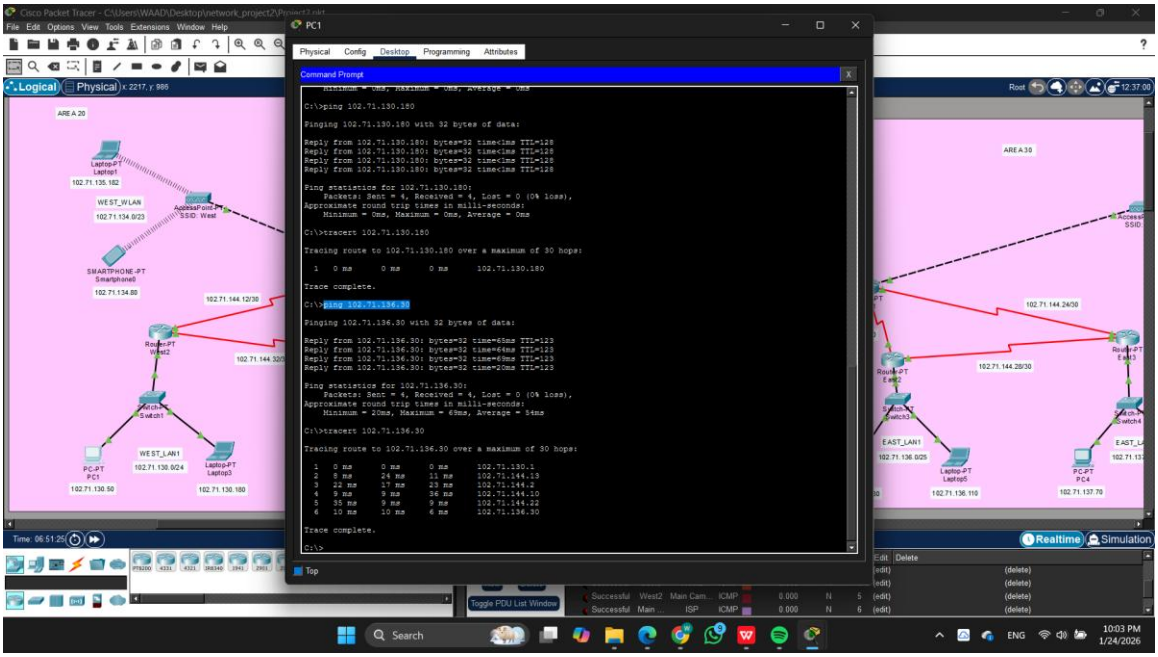


Figure 31 : Inter-area (Area 20 to Area 30)

In this picture, **a device in Area 20 is communicating with a device in Area 30**, so the traffic must travel **between two different areas**. The ping results show **successful replies with 0% packet loss**, which means the routers are correctly routing traffic across areas. The tracert output shows **multiple hops** (more than one router), proving the packets leave the local subnet, go through the **backbone routers**, and then reach the destination network in **Area 30**. This confirms **inter-area routing (Area 20 to Area 30) is working properly**.

10. Inter-area (Area 30 to Area 20)

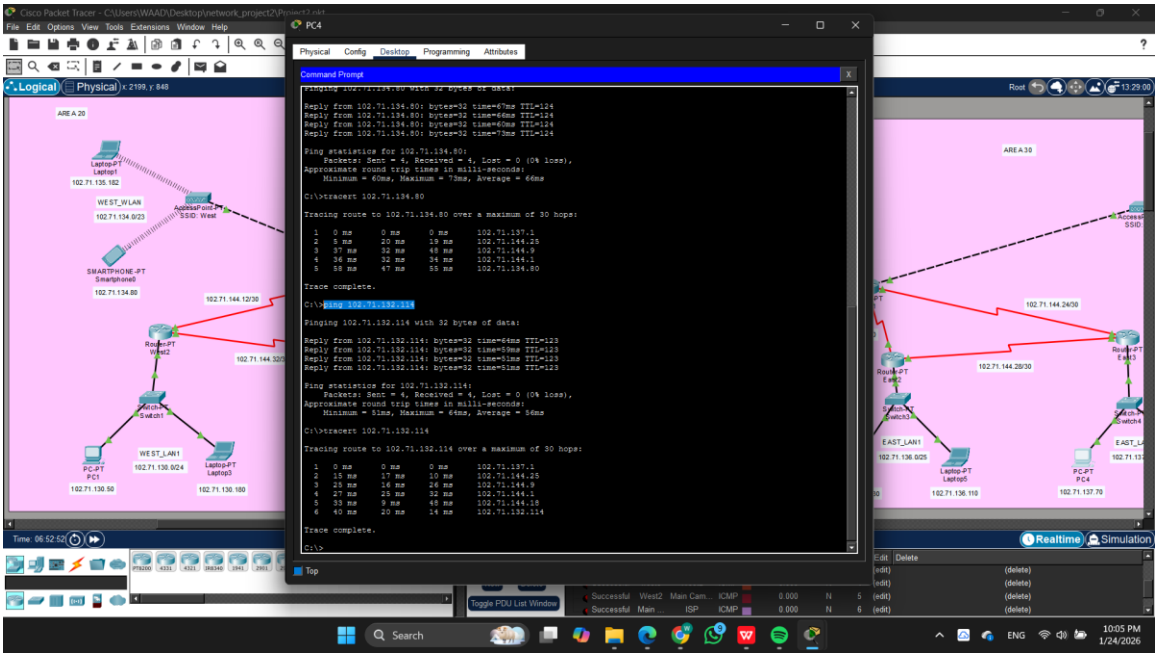


Figure 32 : Inter-area (Area 30 to Area 20)

In this picture, a device **inside Area 30** is testing communication with a device **inside Area 20**, so the traffic must go **between two different areas**. The ping results show **successful replies with 0% packet loss**, which means the destination network in the other area is reachable. The tracert output shows **several hops (multiple routers)**, proving the packets leave Area 30, pass through the **backbone/area border routers**, and then arrive at the subnet in **Area 20**. So the result is: **inter-area routing from Area 30 to Area 20 is working correctly**.

11. To Area 0 (Central LAN) from Area 20

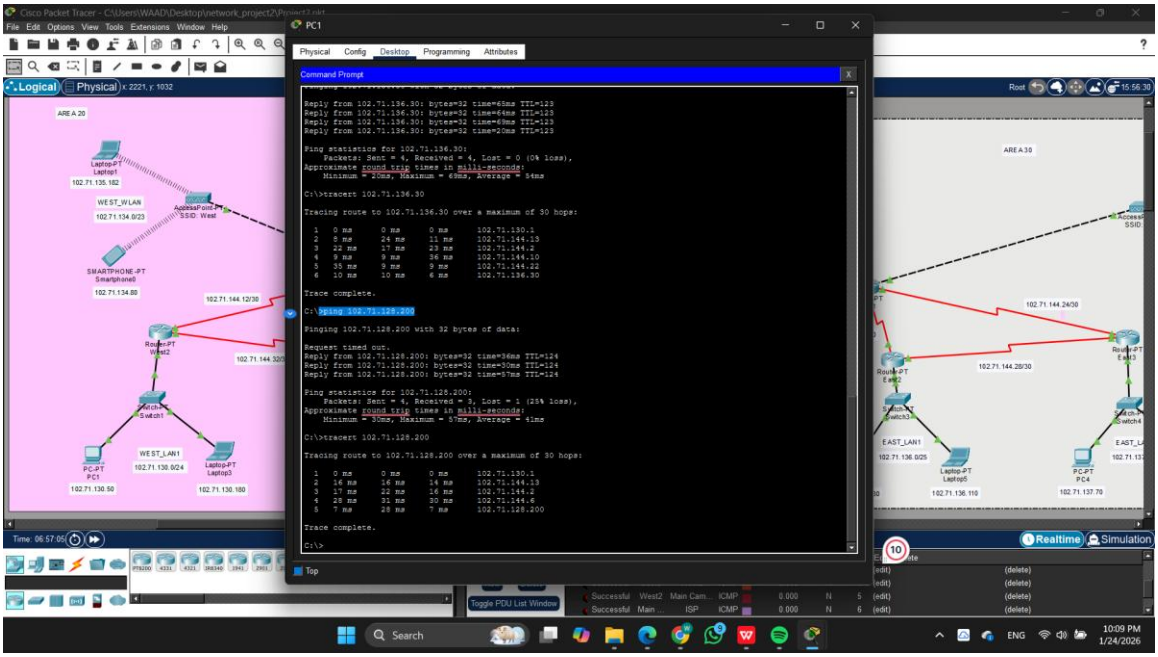


Figure 33 : To Area 0 (Central LAN) from Area 20

In this picture, **PC1 in Area 20 is trying to reach a device in the Central LAN (Area 0)**. The test uses ping and tracert. The ping shows that the destination **is reachable** and the one request timed out at the start is normal. The tracert shows **several hops**, meaning the traffic leaves Area 20, goes through the router to **Area 0**, and finally reaches the Central LAN. So **Area 20 can successfully communicate with Area 0, and inter-area routing to the backbone is working.**

12. To Area 0 (Central LAN) from Area 30

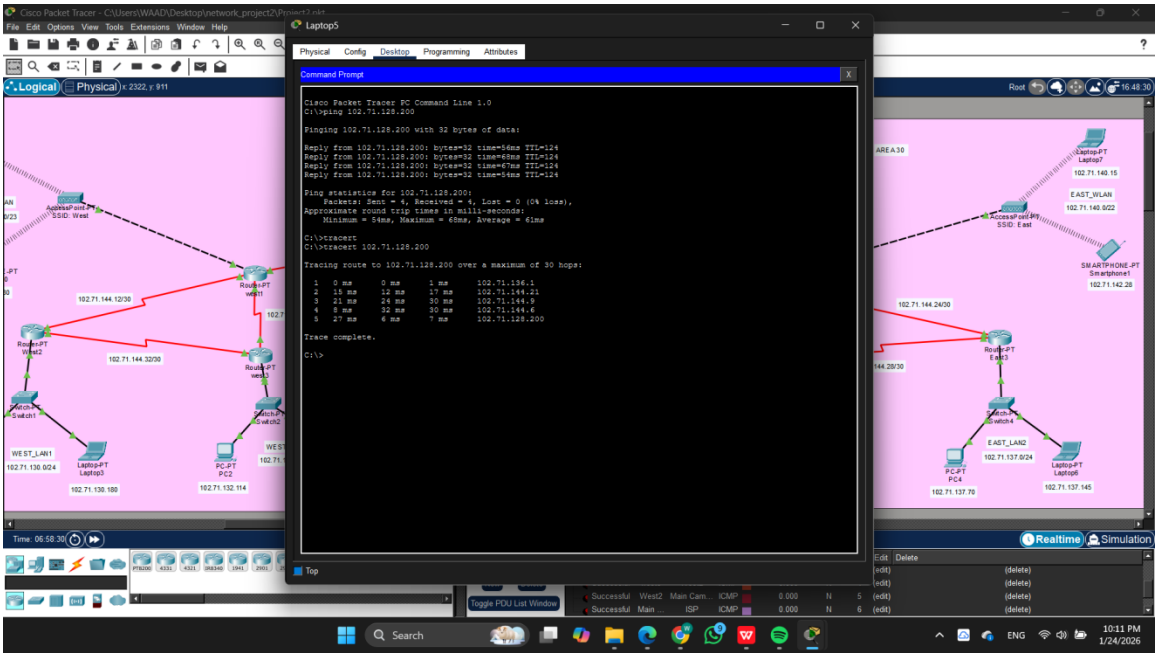


Figure 34 : To Area 0 (Central LAN) from Area 30

In this picture, **Laptop6 (in Area 30)** is testing connectivity to a device in the **Central LAN (Area 0)**. The ping shows **successful replies (no packet loss)**, which means Area 30 can reach the Central network correctly. The tracert shows **multiple hops**, so the packets leave Area 30, go through the routers that connect to the **backbone (Area 0)**, and then arrive at the Central LAN. This confirms that **routing from Area 30 to Area 0 is working**.

13. Wireless to Wired across areas

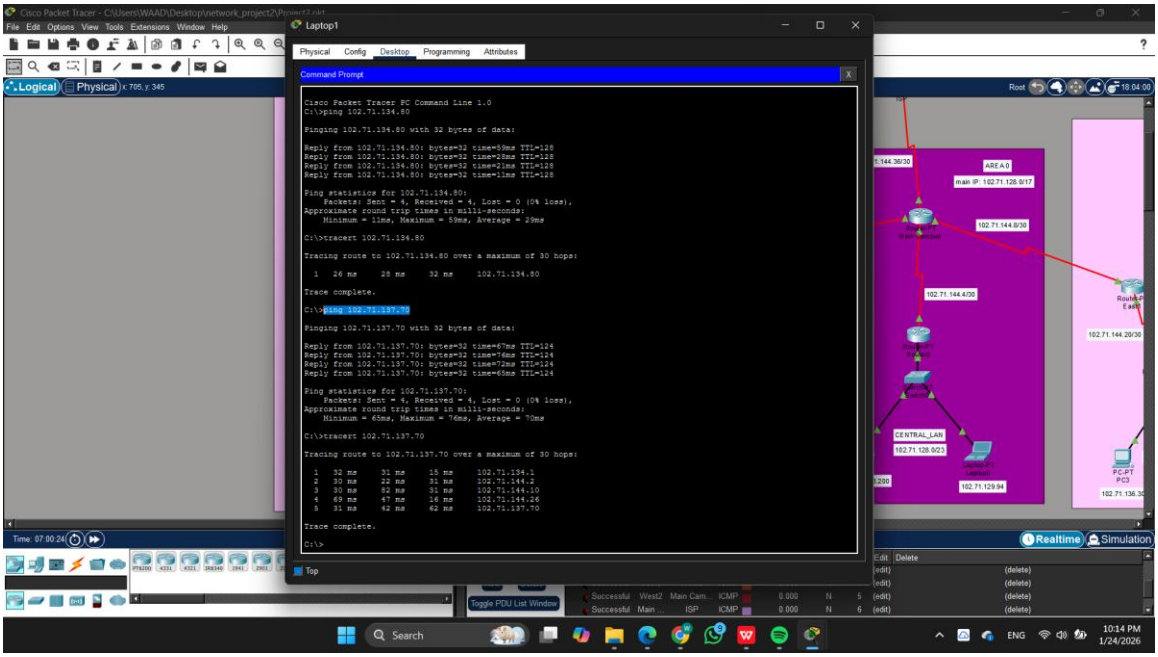


Figure 35 : Wireless to Wired across areas

In this picture, a **wireless device (Laptop1)** is testing communication with a **wired device in a different area**. The ping results show **successful replies with 0% packet loss**, which means the wireless LAN is working, the access point is forwarding traffic correctly, and the routers are routing between areas. The tracert shows **multiple hops**, so the packets go from the laptop to **wireless access point** through the **OSPF routers between areas** and finally to the wired destination. This confirms that **wireless-to-wired communication across different areas is working correctly**.

14. Wireless to Wired across areas

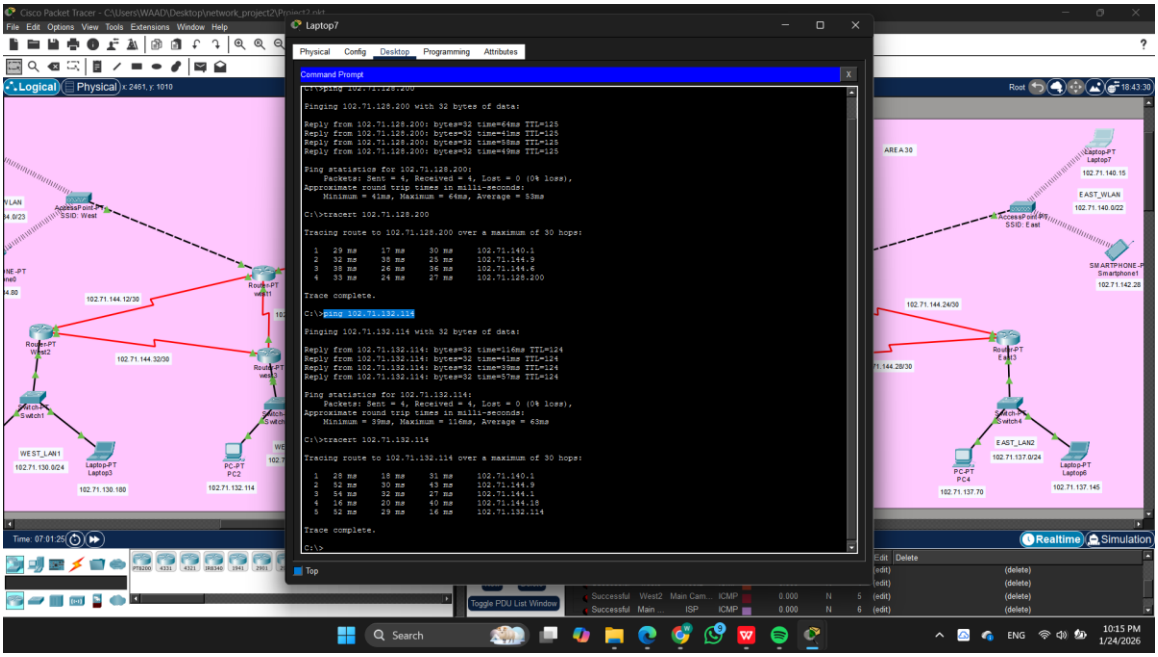


Figure 36 : Wireless to Wired across areas

In this picture, a **wireless device (Laptop7)** is communicating with a **wired device that is located in a different OSPF area**. The ping output shows **successful replies with 0% packet loss**, so the end-to-end connection is working. The tracert shows **multiple hops**, which means the packets go from the laptop through the **wireless network (access point)** to the **default gateway router**, then they travel across the **routers between areas** until they reach the wired destination subnet. This confirms that **wireless-to-wired communication across different areas is configured correctly and working**.

15. Communication with ISP Network (west,east,center)

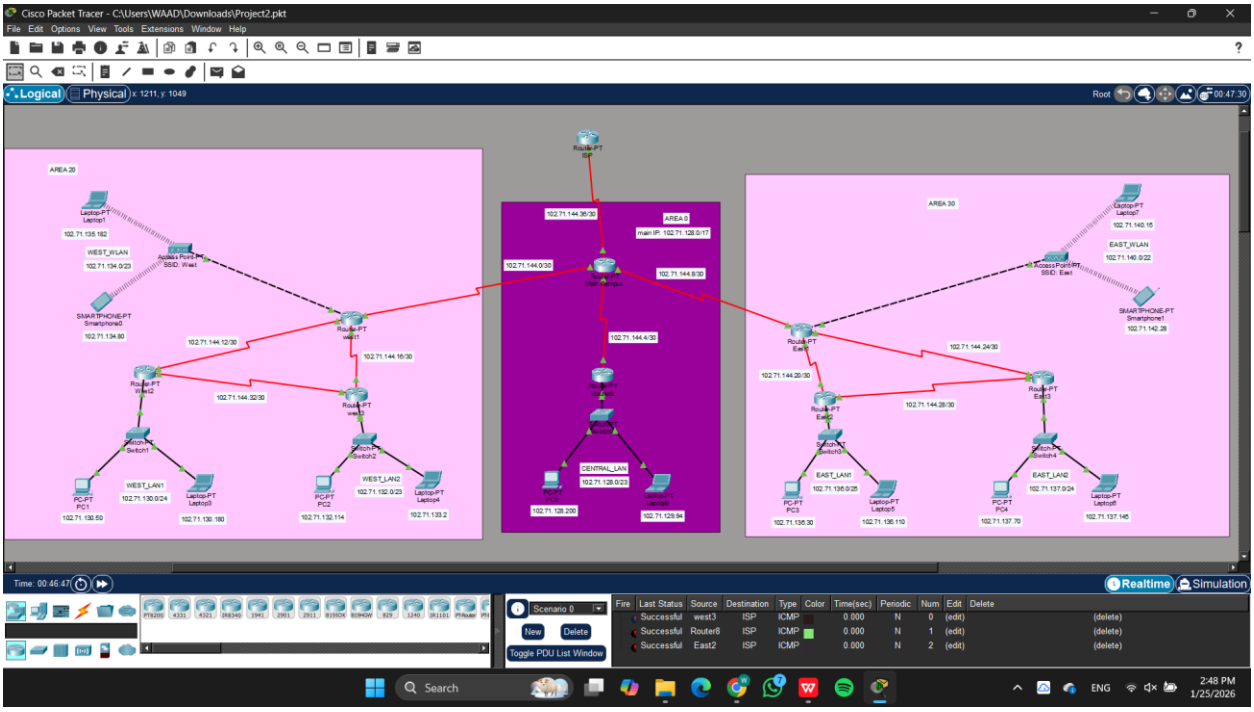


Figure 37 : Communication with ISP Network (west,east,center)

In this test, a packet (message) was sent from the routers in the West, East, and Center areas toward the ISP router using Packet Tracer. The packet successfully moved from each area router, through the internal links and then reached the ISP side. This shows that the router-to-ISP connection is working, the WAN interfaces are up, and the routers have the correct routing path (they know how to forward traffic toward the ISP from every area).

Discussion:

Successful pings across different areas demonstrate that inter-area routing is functioning correctly and that default gateways on end devices are properly configured. Tracert outputs provide additional confidence by revealing the sequence of routers used.

Routing verification (show ip route)

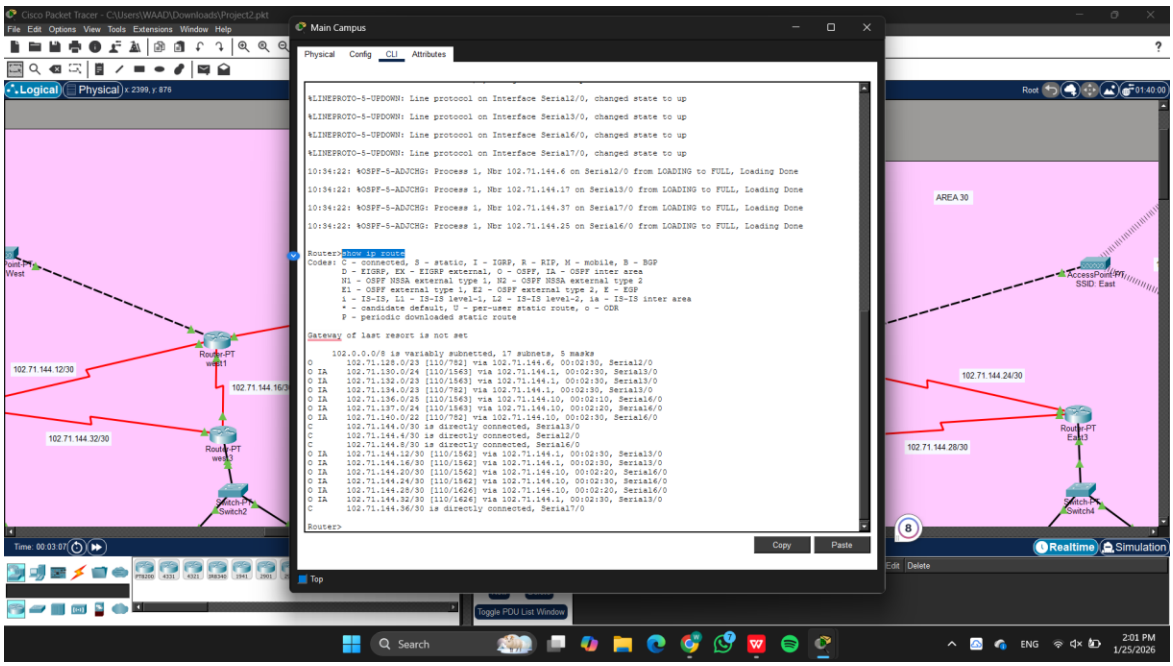


Figure 38 : Routing verification show ip route Main Campus

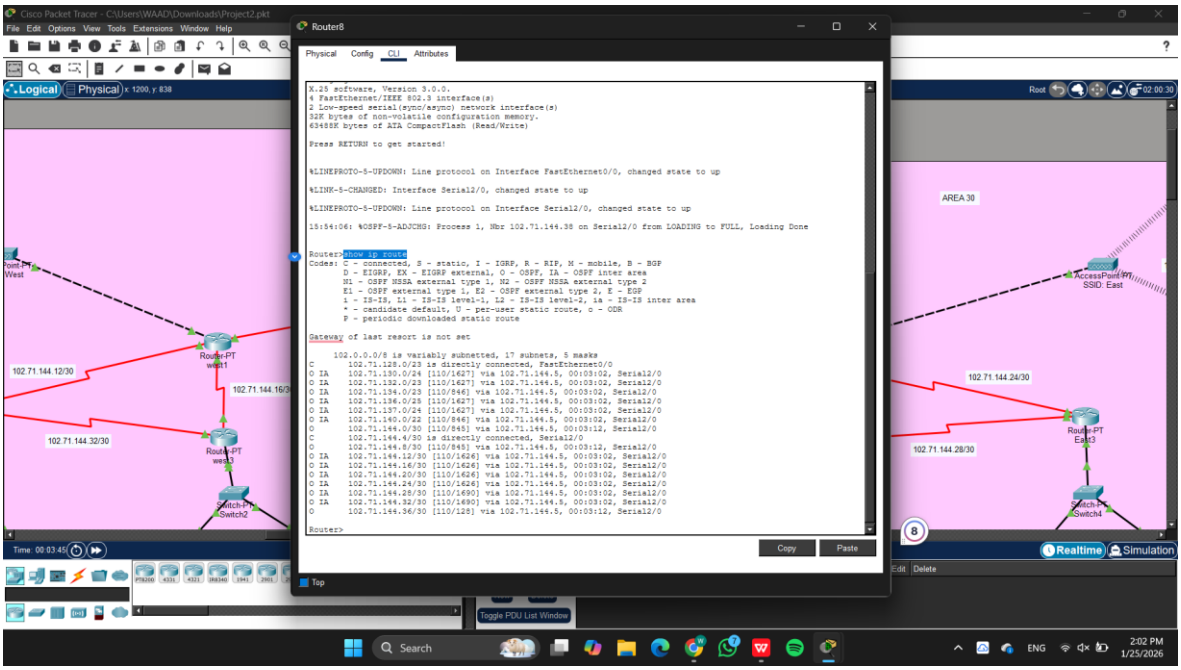


Figure 39 : Routing verification show ip route Router8

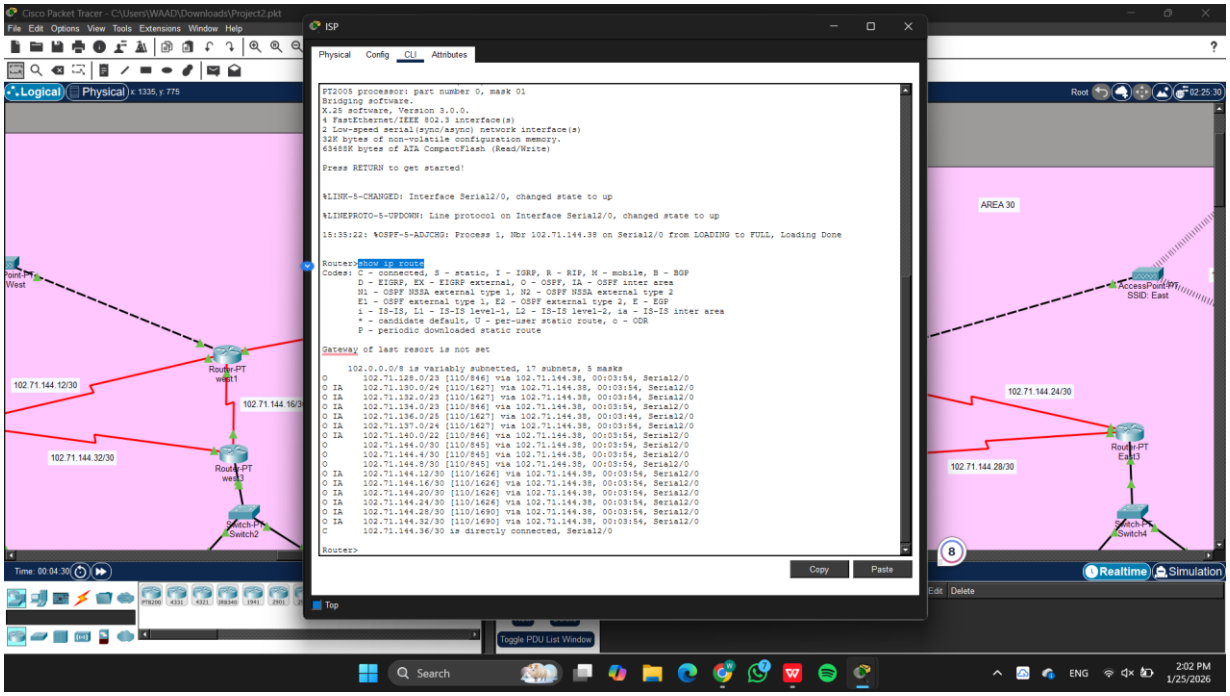


Figure 40 : Routing verification show ip route ISP

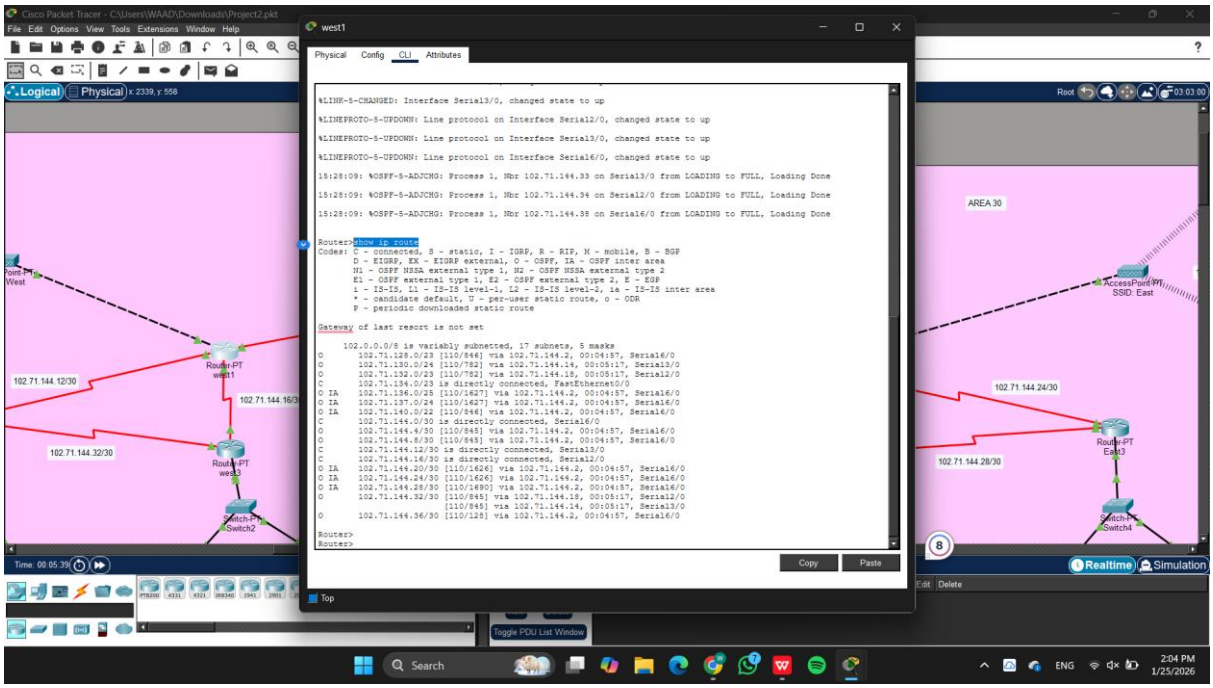


Figure 41 : Routing verification show ip route West1

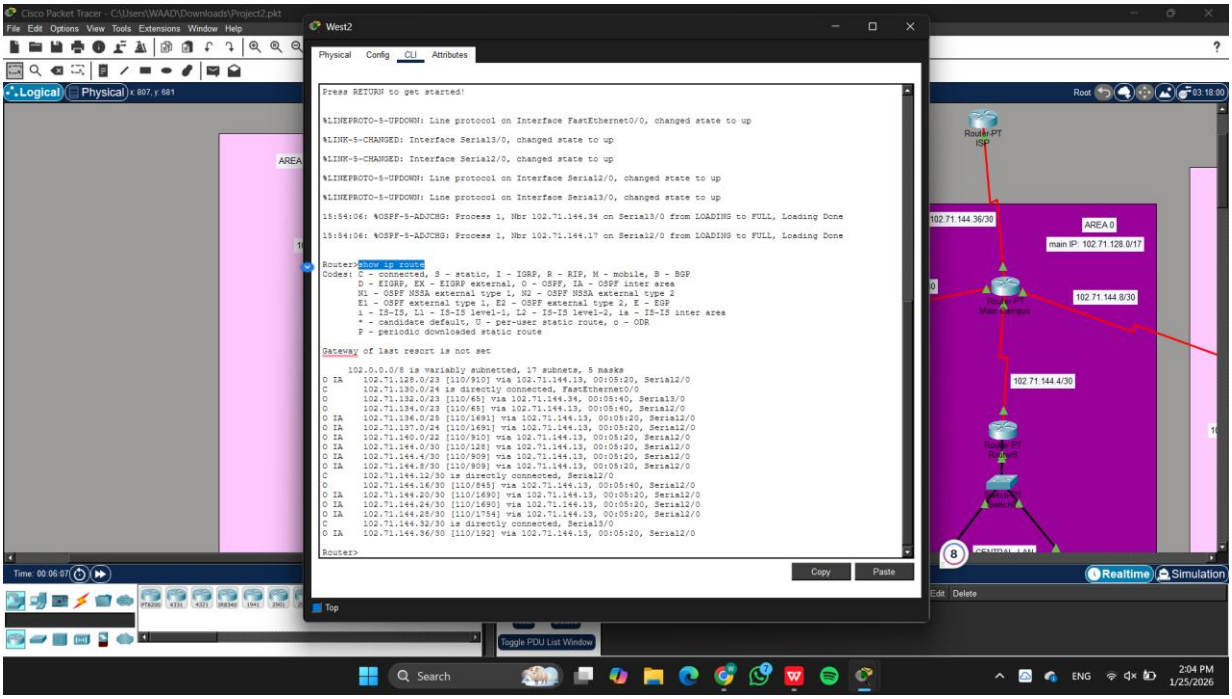


Figure 42 :Routing verification show ip route West2

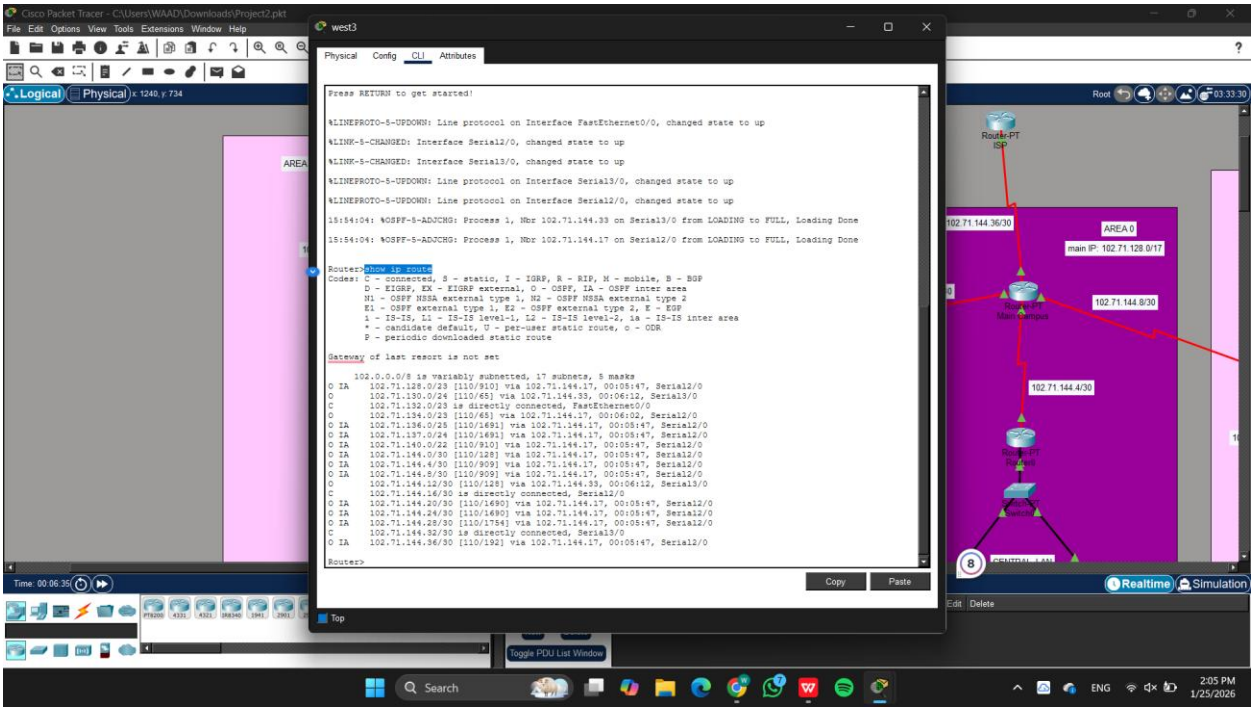


Figure 43 : Routing verification show ip route West3

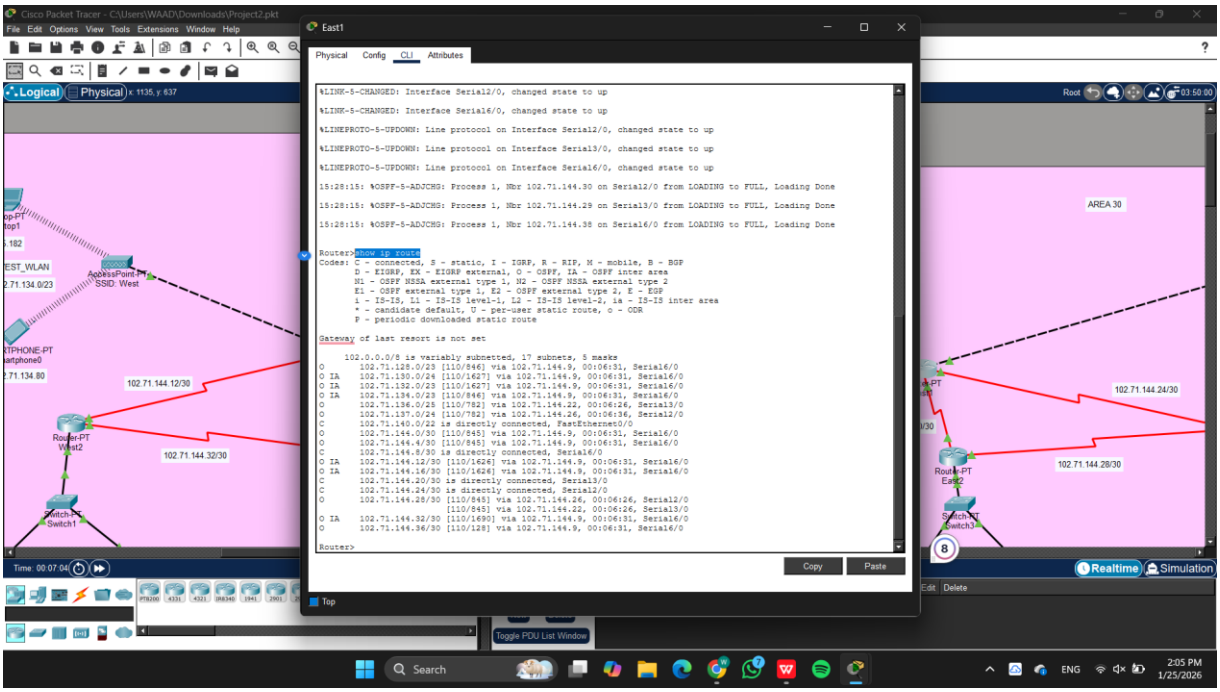


Figure 44 :Routing verification show ip route East1

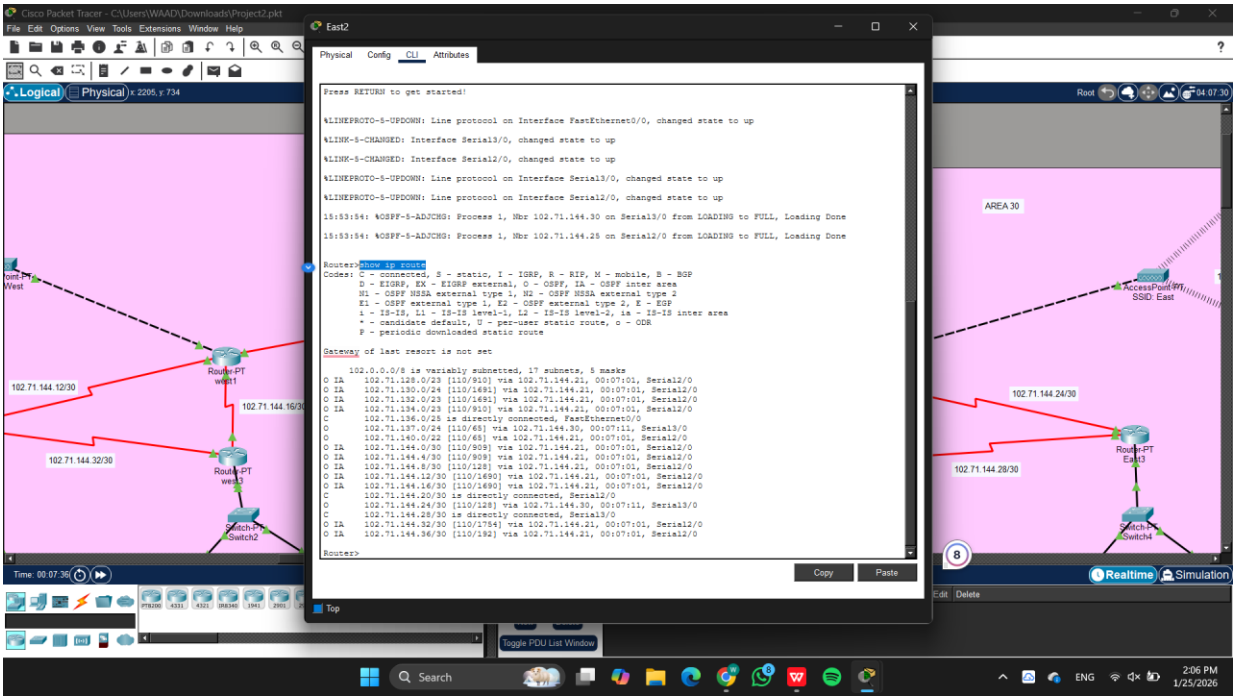


Figure 45 : Routing verification show ip route East2

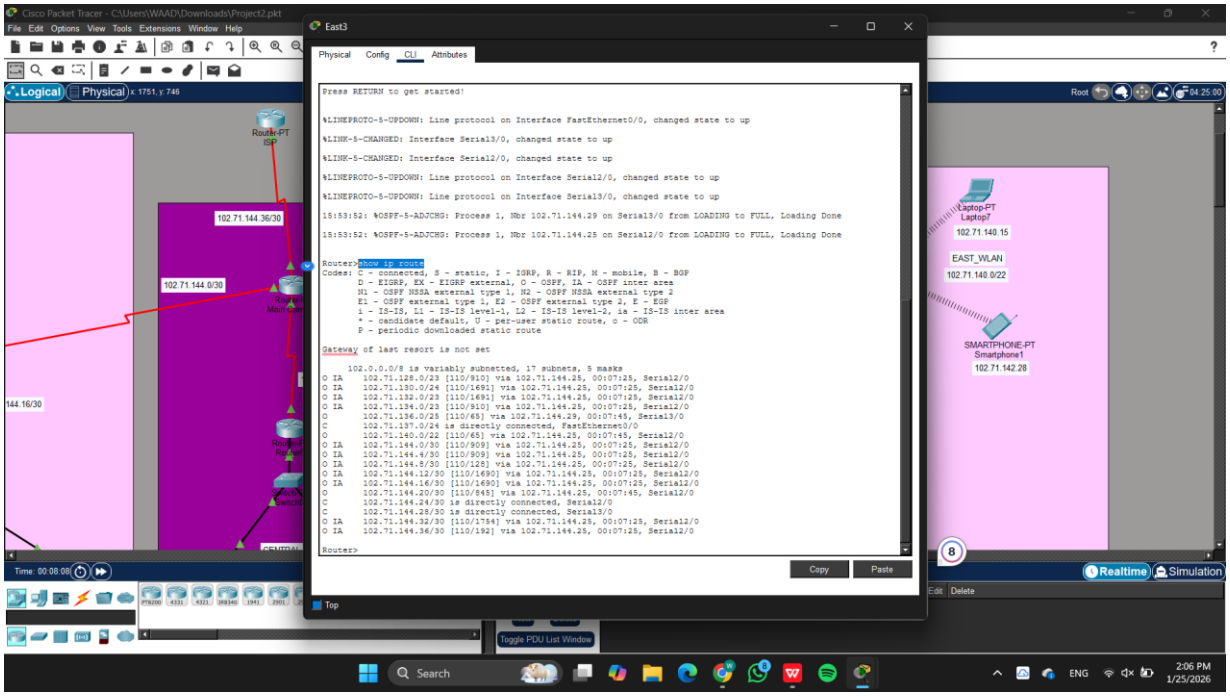


Figure 46 : Routing verification show ip route East3

Summary of results

All required tasks were implemented and validated: topology labeling, static IP addressing for routers and hosts, multi-area routing, and end-to-end connectivity across wired and wireless segments. The evidence provided in Figures and Tables demonstrates that the network meets the project constraints and supports reliable communication between all end devices.

Issues and Limitations

During the implementation of this project, several issues and limitations were encountered that affected the development and testing process. One of the main challenges was ensuring accuracy in IP addressing and subnetting using VLSM, since even a small mistake in selecting the prefix length or defining address ranges could lead to incorrect interface configurations and result in connectivity failures between different networks. In addition, configuring multi-area OSPF required careful attention to area assignments and proper router connections, as any misconfiguration could prevent routes from being advertised and exchanged correctly between areas. Furthermore, the wireless network configuration introduced additional challenges, especially in setting WPA2 security parameters and verifying wireless client connectivity, where incorrect password or encryption settings could cause unstable connections or complete connection failure.

Moreover, troubleshooting required repeated testing using commands such as ping, tracert, and show ip route, which increased the time and effort needed, particularly when problems occurred across multiple routers. From a teamwork perspective, time management and coordination also presented a limitation, as it was difficult to find a suitable shared time for all team members to work together simultaneously. As a result, tasks were sometimes divided and completed individually, then combined later to produce the final project configuration.

Team work

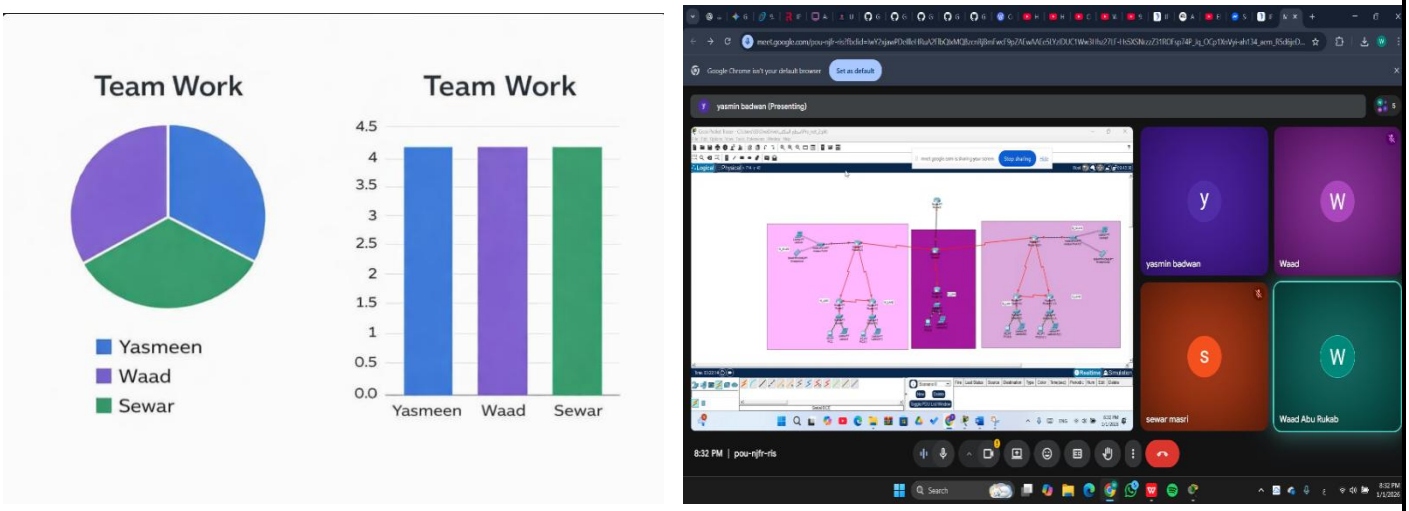


Figure 47:Contribution distribution among team members

Throughout this project, our team of three students — **Yasmeen, waad, and Sewar** — maintained strong collaboration and effective communication. We worked together closely during all stages of the project, including planning, designing, implementing, testing, and documenting the network topology in Cisco Packet Tracer. Regular meetings and discussions were held to divide tasks clearly and ensure that all project requirements were achieved accurately.

Each team member contributed equally to the completion of the project. Responsibilities were distributed among **subnetting and IP addressing**, configuring network devices such as **routers, switches, and access points**, implementing **OSPF multi-area routing**, and setting up **wireless networks with WPA2 security**. In addition, the team collaboratively performed connectivity verification using tools such as **ping and traceroute**, and prepared the final documentation and figures required in the report.

Our workflow was highly cooperative and interactive. Team members frequently reviewed each other's configurations, cross-checked addressing tables, verified routing tables, and assisted in troubleshooting issues related to OSPF neighbor formation, routing advertisements, IP conflicts, or wireless connectivity. This teamwork helped us detect errors early and ensure the network operated efficiently and met all required specifications.

Beyond the technical achievements, this project significantly strengthened our teamwork and communication skills. Working as a group taught us how to coordinate tasks effectively, manage responsibilities under time constraints, and solve problems collaboratively. Overall, this experience improved our confidence and ability to work efficiently in team-based networking and communication engineering projects.

Conclusion

In conclusion, this project successfully achieved the required network design, configuration, and verification tasks. A structured IPv4 addressing plan was developed from the main network 102.71.128.0/17 and efficiently subnetted using VLSM to allocate suitable address spaces for all LANs and WLANs. Router-to-router links were implemented using /30 subnets to reduce IP waste while maintaining clear point-to-point connectivity. OSPF was configured as a multi-area routing protocol with Area 0 as the backbone and Areas 20 and 30 for regional segmentation, ensuring scalable and organized routing across the entire topology. Comprehensive testing using ping, traceroute, and routing table validation confirmed correct addressing, proper gateway configuration, stable OSPF adjacencies, and successful communication between all wired and wireless end devices. Overall, the implemented network satisfies the project constraints and demonstrates effective use of subnetting and dynamic routing principles in a realistic campus network environment

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[3] Types of Network Topology - GeeksforGeeks(Accessed on 24/1/2026)

[4] Open Shortest Path First (OSPF) Protocol Fundamentals - GeeksforGeeks(Accessed on 24/1/2026)

[5] WLAN Full Form - Wireless Local Area Network - GeeksforGeeks(Accessed on 24/1/2026)

[6] Routing - GeeksforGeeks(Accessed on 24/1/2026)

[7] Network Devices - GeeksforGeeks(Accessed on 24/1/2026)